



Aras Innovator 33

Microsoft Reporting Services Guide

For use with Microsoft SQL Server 2019

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Aras Corporation
100 Brickstone Square
Suite 100
Andover, MA 01810
Phone: 978-691-8900

E-mail: support@aras.com

Website: <https://www.aras.com/>

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1 Introduction

Aras Innovator's Report ItemType includes support for Microsoft's 'Reporting Services' in addition to supporting the existing XSLT based reports.

Reporting Services is an integral component of SQL Server 2019, and any customer with a SQL Server 2019 license can install Reporting Services. (See installation instructions below).

Usage of this release is recommended only for those with advanced Aras Innovator and SQL training.

Primary benefits to switching from stylesheet driven reports to Report Server:

- Query building is in SQL and is graphical (same as in Enterprise Manager).
- Report definitions are transportable, you can develop on your laptop, and then deploy to a production server.
- Report layout is more drag-n-drop and has more powerful formatting tools.
- The resulting reports are paginated.
- The end-user can easily print, PDF, or export to Excel the report output.
- Excel like Charts and graphs can be produced

2 Installing Reporting Services

Reporting Services is a component of Microsoft SQL server 2019. Click on this link to download Reporting Services separately: <https://www.microsoft.com/en-us/download/details.aspx?id=100122>. The Microsoft Download page appears.

Use the following procedure to download Microsoft SQL Server 2019 Reporting Services and install it:

1. Scroll down to **Microsoft SQL Server 2019 Reporting Services**.

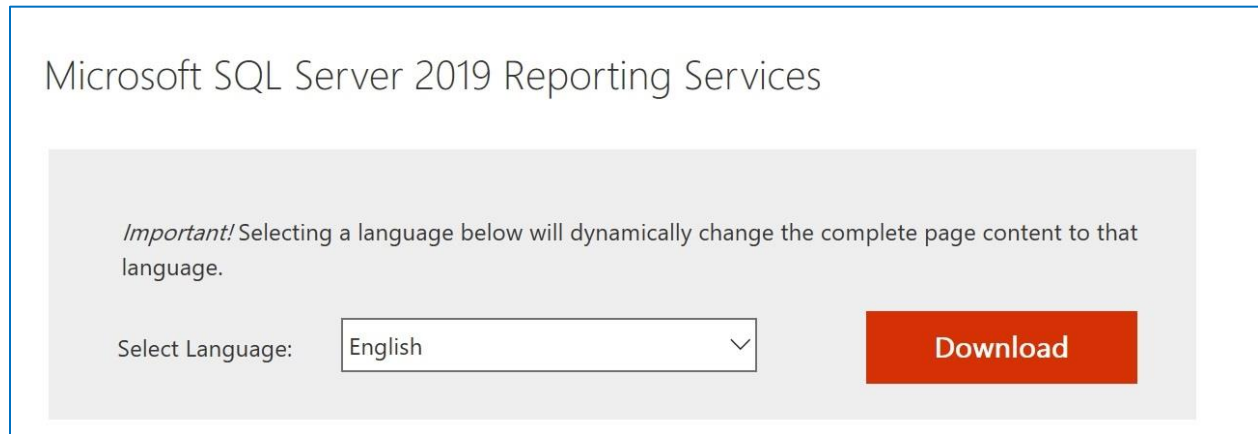


Figure 1.

2. Click **Download**. The dialog box shown in Figure 2 appears.

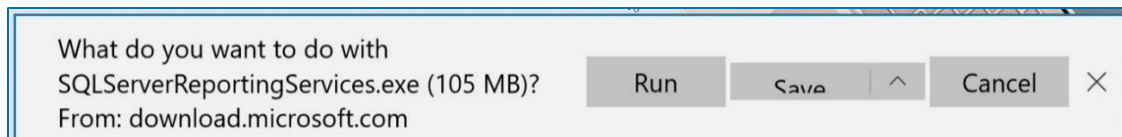


Figure 2.

3. Click **Run**. A security scan is run. When that is successfully completed The dialog box shown in Figure 3 appears.



Figure 3.

4. Click **Install Reporting Services**. The dialog box shown in Figure 4 appears.



Figure 4.

5. Select the **Enter the product key** button and type the product key in the appropriate field.

6. Click **Next**. The **Review the license terms** dialog appears.



Figure 5.

7. Select **I accept the license terms** and click **Next**. The **Install Database Engine** dialog appears.



Figure 6.

- Click **Next**. The **Specify an install location** dialog appears.



Figure 7.

- Use the default location shown here or click **Browse** to select a different install location.
- Click **Install**. When the set-up procedure is complete, the dialog box shown in Figure 8 appears.



Figure 8.

- Click **Configure report server**. The **Report Server Configuration Manager** dialog appears.

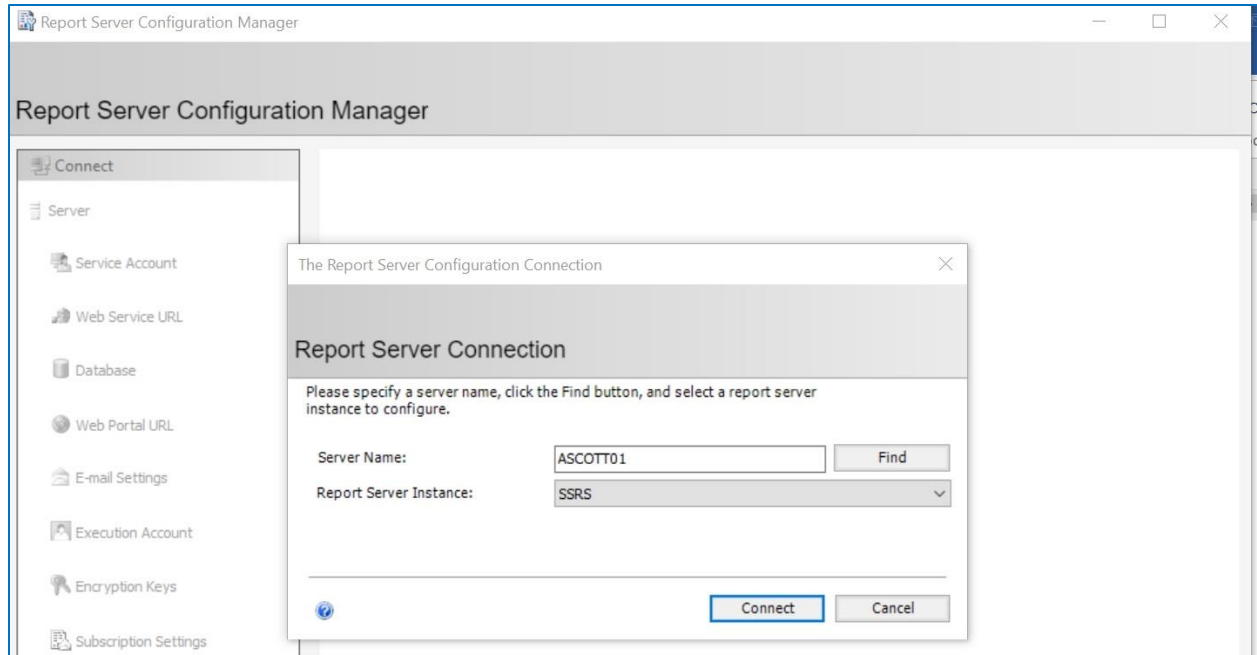


Figure 9.

12. Click **Find** to select a different server name.
13. Click the **down arrow** in the **Report Server Instance** field to select a different Report Server Instance.

14. Click **Connect** to connect to the appropriate server. The dialog box shown in Figure 10 appears.

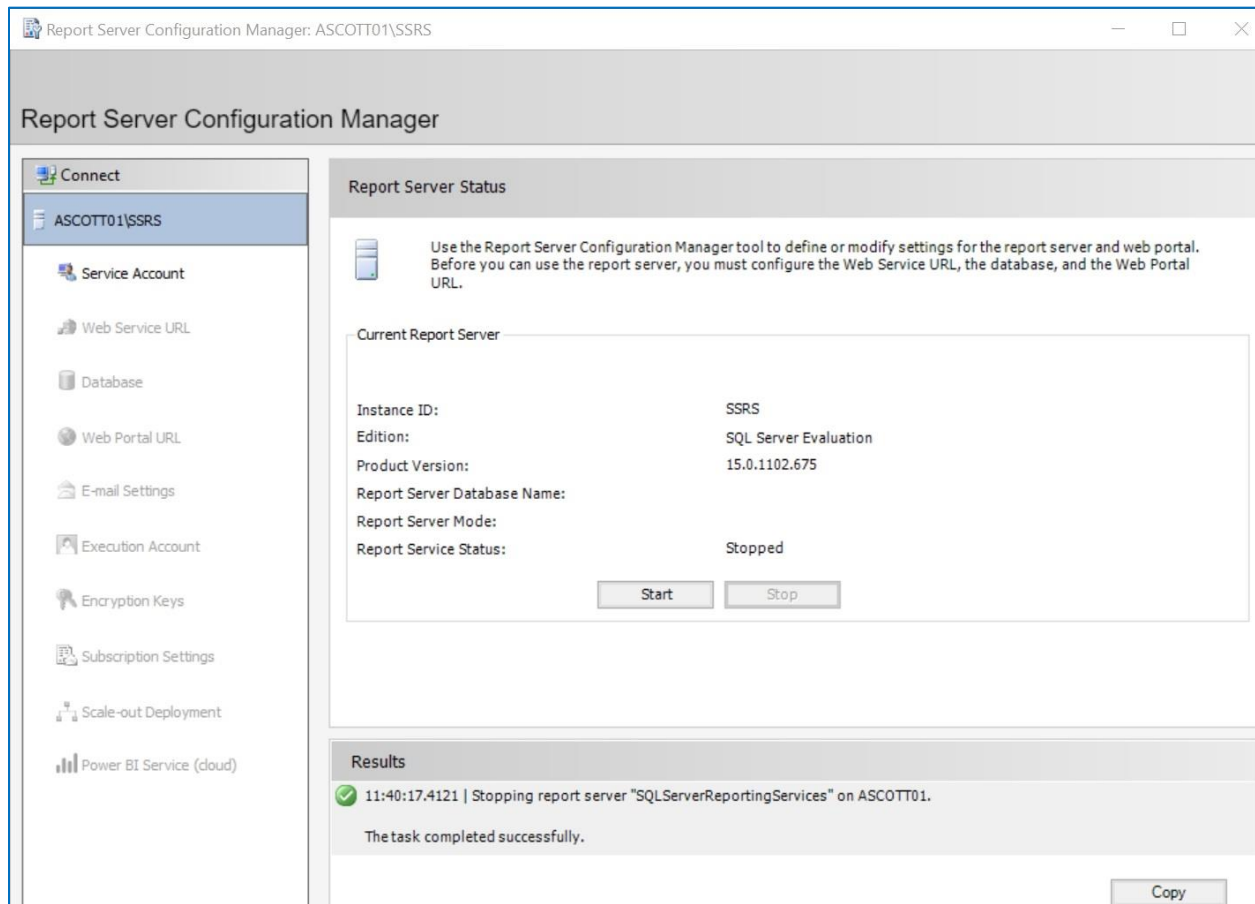


Figure 10.

15. Click **Start**. You can modify or define settings for the Report Server. Click **Stop** to stop the Report Server.

In addition to selecting the Reporting Services option, you can select **Analysis Services** that provides graphing and charting functionality.

2.1 Installation is a component of SQL Server 2019

This section is a summary of dialogs for installing SQL server 2019.

Note: If you view dialog boxes not listed here, use the default selections.

2.1.1 Default Instance

SQL Server 2019 Setup

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Product Key

License Terms

Global Rules

Microsoft Update

Product Updates

Install Setup Files

Install Rules

Feature Selection

Feature Rules

Instance Configuration

Server Configuration

Database Engine Configuration

Feature Configuration Rules

Ready to Install

Installation Progress

Complete

Default instance

Named instance: MSSQLSERVER

Instance ID: MSSQLSERVER

SQL Server directory: E:\Program Files\Microsoft SQL Server\MSSQL15.MSSQLSERVER

Installed instances:

Instance Name	Instance ID	Features	Edition	Version
---------------	-------------	----------	---------	---------

< Back

Next >

Cancel

Figure 11.

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13

2.1.2 Service account Setup

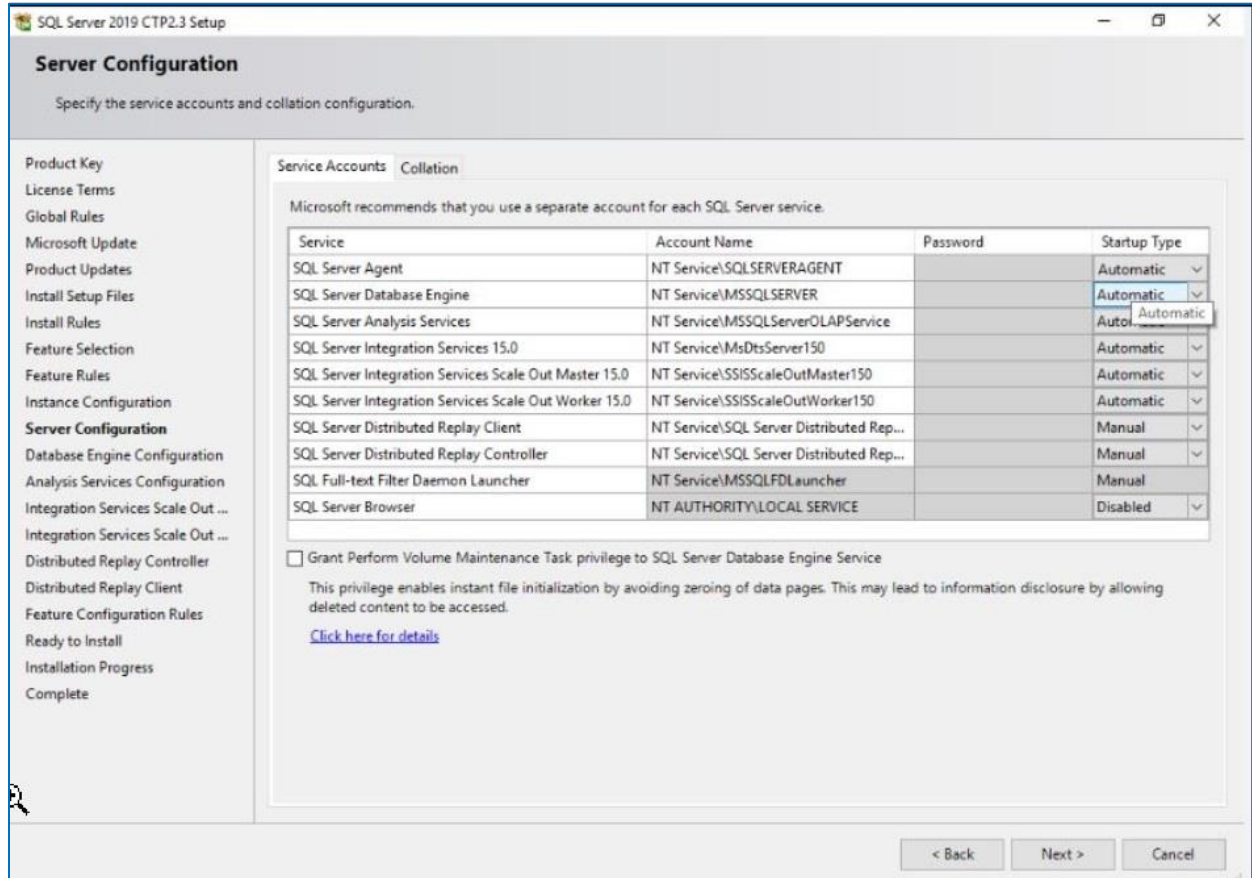


Figure 12.

2.1.3 Database Engine Configuration and Authentication Mode

SQL Server 2019 Setup

Database Engine Configuration

Specify Database Engine authentication security mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and Filestream settings.

License Terms
Global Rules
Product Updates
Install Setup Files
Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Installation Progress
Complete

Server Configuration | Data Directories | TempDB | Memory | User Instances | FILESTREAM

Specify the authentication mode and administrators for the Database Engine.

Authentication Mode

☐ Windows authentication mode

☒ Mixed Mode (SQL Server authentication and Windows authentication)

Specify the password for the SQL Server system administrator (sa) account.

Enter password: [Masked Password]

Confirm password: [Masked Password]

Specify SQL Server administrators

[Add Current User] [Add...] [Remove]

SQL Server administrators have unrestricted access to the Database Engine.

< Back | Next > | Cancel

Figure 13.

2.1.4 Analysis Services Configuration

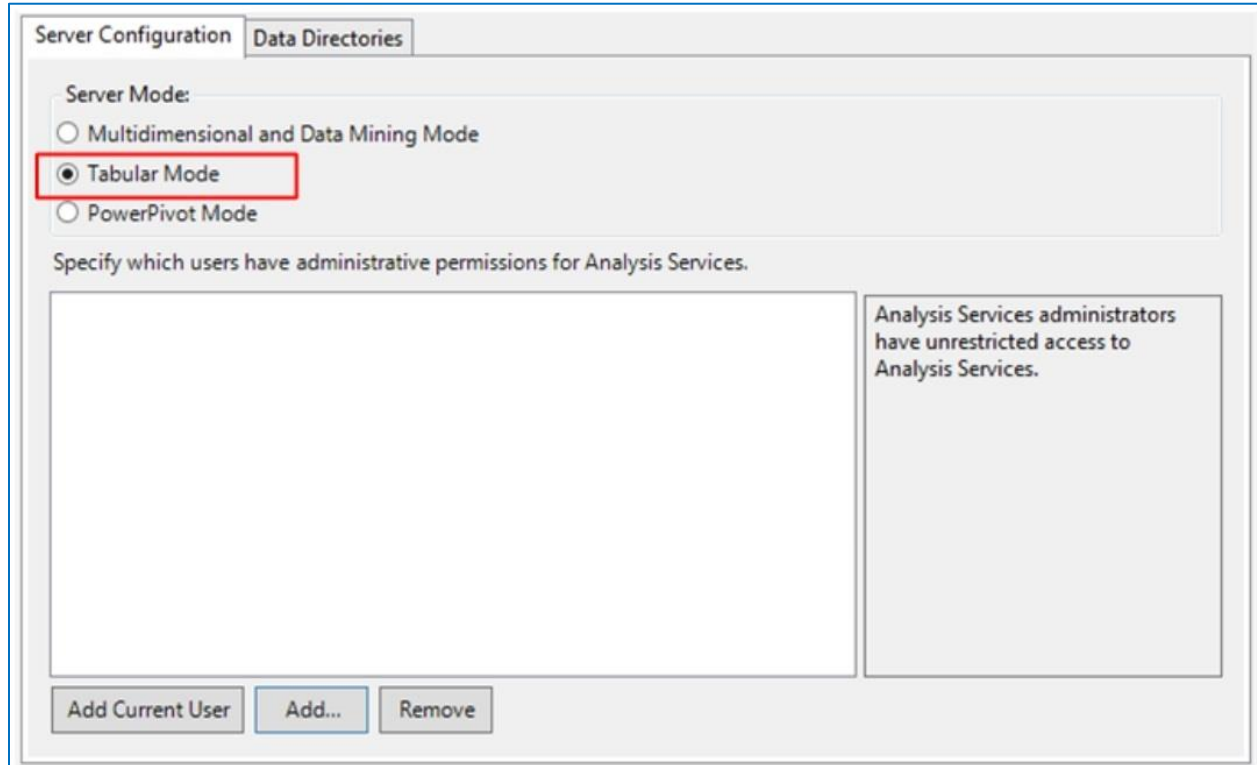


Figure 14.

2.1.5 Reporting Services Options

Reporting Services Configuration
Specify the Reporting Services configuration mode.

Global Rules
Product Updates
Install Setup Files
Install Rules
Installation Type
Product Key
License Terms
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Analysis Services Configuration
Reporting Services Configura...
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

Reporting Services Native Mode

☒ Install and configure.
Installs and configures the report server in native mode. The report server is operational after setup completes.

☐ Install only.
Installs the report server files. After installation, use Reporting Services Configuration Manager to configure the report server for native mode.

Reporting Services SharePoint Integrated Mode

☐ Install only.
Installs the report server files. After installation use SharePoint Central Administration to complete the configuration. Verify the SQL Server Reporting Services service is started and create at least one SQL Server Reporting Services service application. For more information, click Help.

< Back Next > Cancel

Figure 15.

2.1.6 Remaining Selections

The default selections can be used from this point forward.

3 Configuring Reporting Services

The Reporting Services use Folders to organize Reports. The Aras Innovator integration with Reporting Services use as the default **Innovator Reports** Folder.

In this section, we configure the Reporting Services using the Report Manager by adding the **Innovator Reports** Folder and a Shared Data Source, which the Reports use to connect to the database.

3.1 Installing Reporting Services

The **ReportServer** Reporting Services comes with the **Report Manager** page used to configure the Reports, Folders, Data Sources, and so on. This page has been integrated into Innovator client via an Innovator Report that opens the **Report Manager** page.

Open your browser with <http://localhost:80/Reports> on the default installation.

To set up Reporting Services:

1. Open the Reporting Services Configuration Manager.
2. Set up a Service Account.

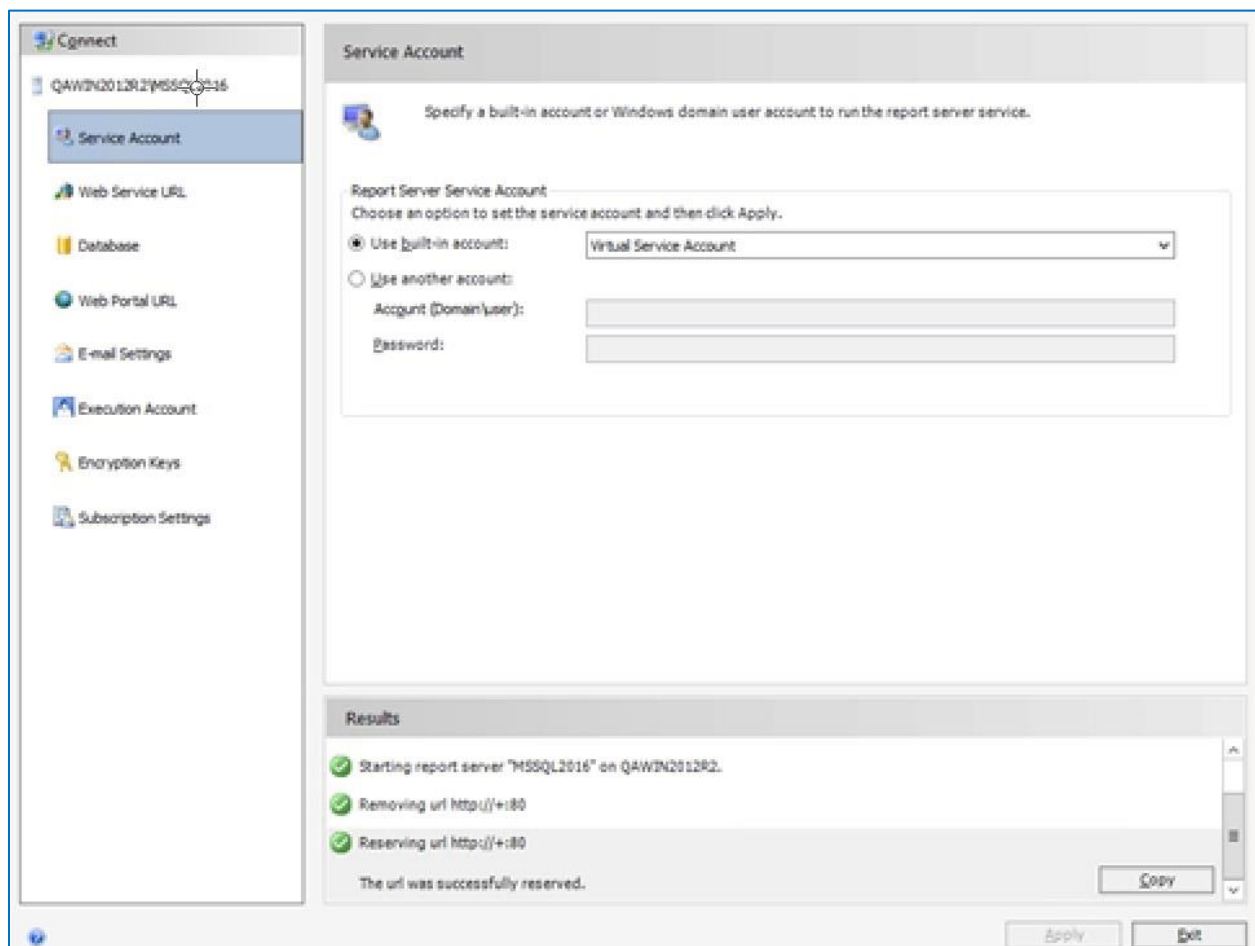


Figure 16.

3. Set up a Web Service URL

The screenshot displays the 'Web Service URL' configuration window. On the left is a sidebar with a 'Connect' icon and a list of settings: 'QAWIN2012R2/MSQR/2016', 'Service Account', 'Web Service URL' (selected), 'Database', 'Web Portal URL', 'E-mail Settings', 'Execution Account', 'Encryption Keys', and 'Subscription Settings'. The main panel is titled 'Web Service URL' and contains the following sections:

- Report Server Web Service Virtual Directory:** A text box labeled 'Virtual Directory:' containing the value 'ReportServer2016'.
- Report Server Web Service Site identification:** A section with four fields:
 - IP Address:** A dropdown menu showing 'All Assigned (Recommended)'.
 - TCP Port:** A text box containing '80'.
 - HTTPS Certificate:** A dropdown menu showing '(Not Selected)'.
 - HTTPS Port:** An empty text box.
- Report Server Web Service URLs:** A section with a text box labeled 'URL(s):' containing 'http://QAWIN2012R2/MSQR/ReportServer2016'.

At the bottom right of the main panel is an 'Advanced...' button. Below the main panel is a 'Results' section, which is currently empty. At the bottom right of the window are 'Copy', 'Apply', and 'Exit' buttons.

Figure 17.

4. Set up the Database.

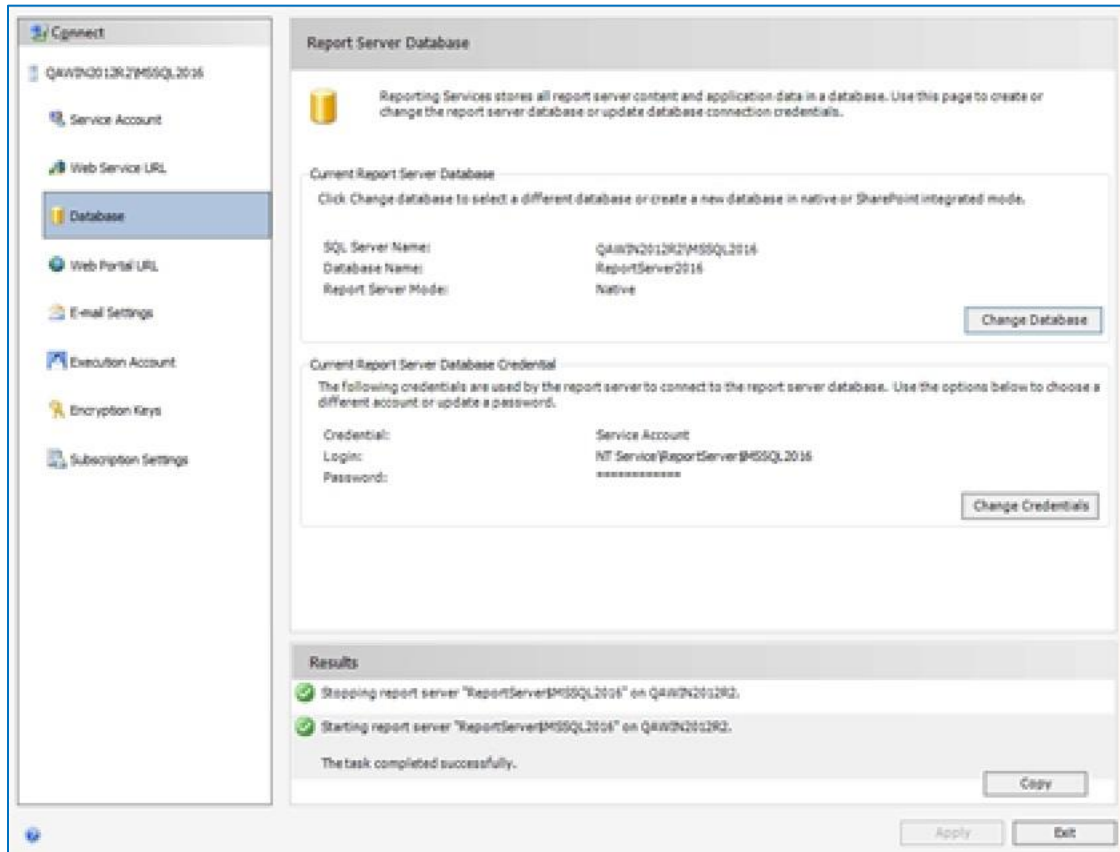


Figure 18.

5. Set up the Web Portal URL.

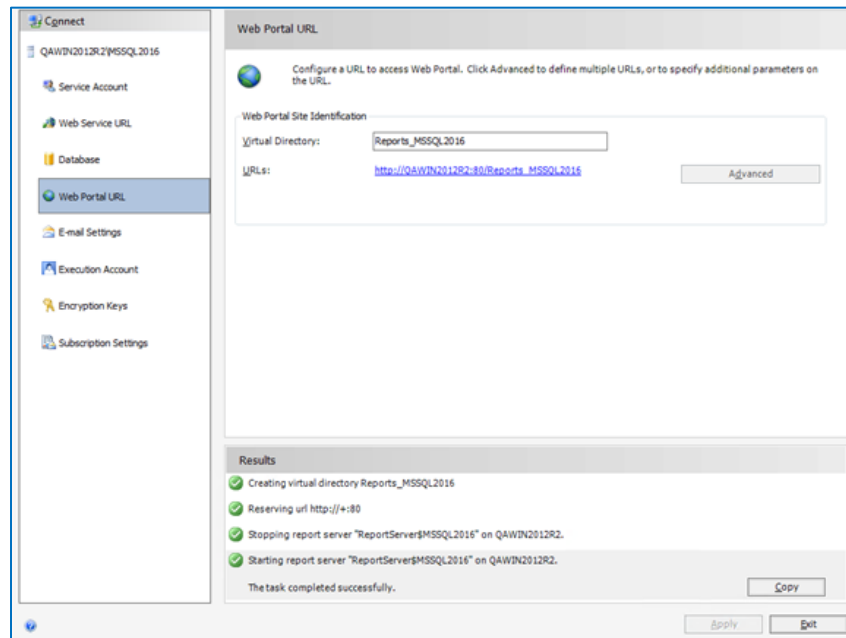


Figure 19.

3.2 Setting Up the Report Folder

1. Open the **Report Manager** page (defined above) in a new dialog.

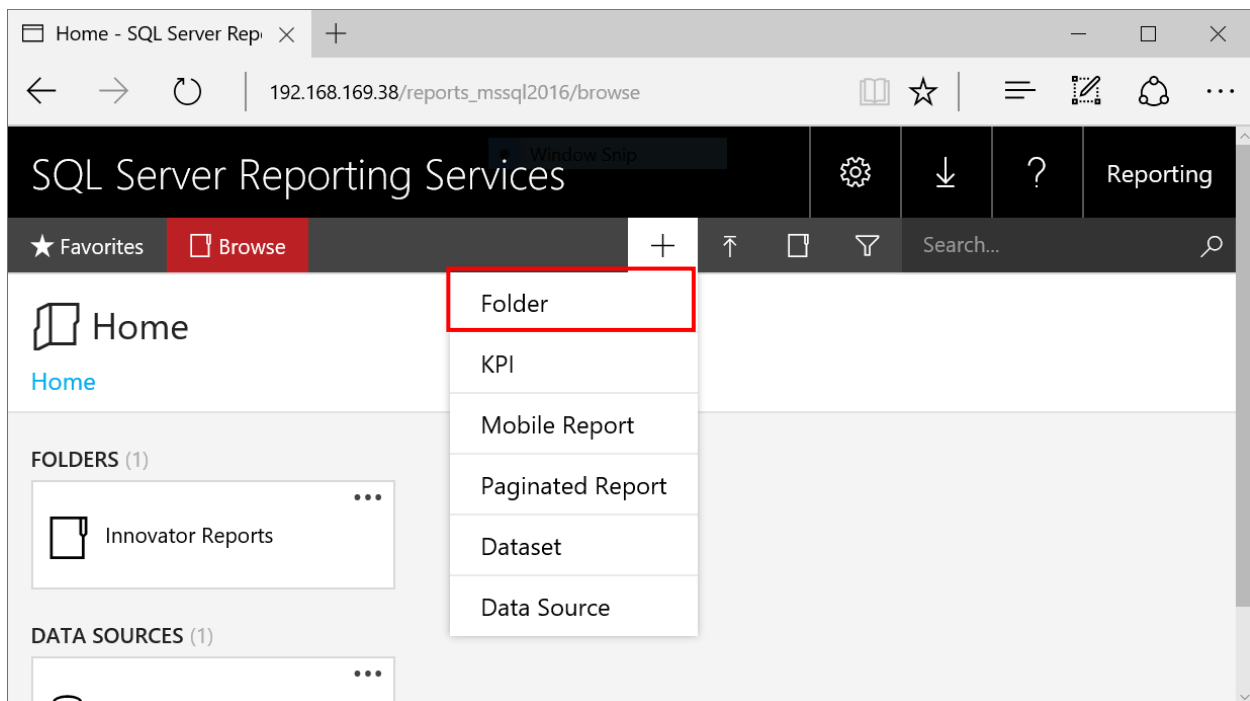


Figure 20.

2. Click **+** and then **Folder**.

3. Enter **Innovator Reports** for the name and click **Create**.

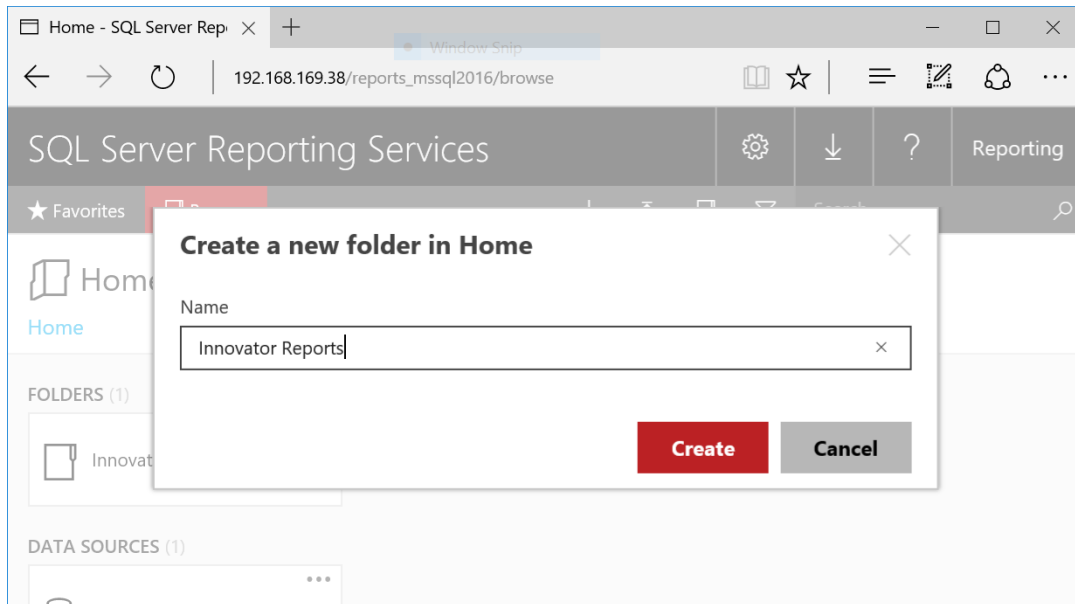


Figure 21.

Note: If you use a folder named something other than **Innovator Reports**, you need to change the **Report Server Folder** variable in the database as well in TOC --> **Administration** --> **Variables**. The default value for this is **Innovator Reports**, but it can be customized for each database. For the purposes of these instructions, we assume the default value is used.

3.3 Adding a Shared Data Source

Once you add the **Innovator Reports** Folder, navigate into it by clicking on the link. Now, you add a Shared Data Source for the Reports to use to connect to the database.

1. Click **+** and then **Data Source** inside the **Innovator Reports** folder.

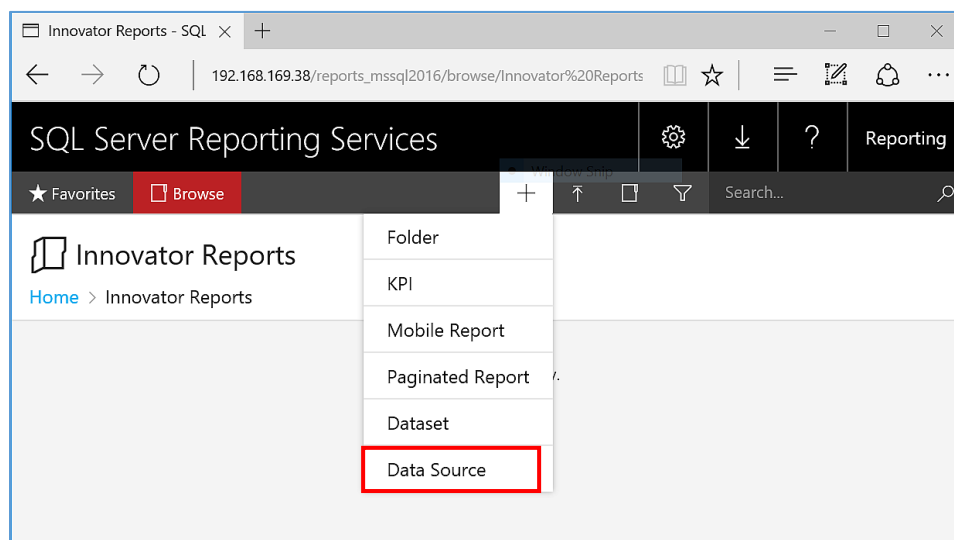


Figure 22.

The **New data source** dialog appears.

Figure 23.

2. Enter **innovator_regular** for the Data Source Name.

Note: By using this same name for both the development and production servers, exporting and importing Reports becomes simpler because there is no need to reassign the Data Source for the Reports you import.

3. Enter the following for the Connection String replacing the values for the two parameters with the actual server name where SQL Server is installed as well as the actual database name.
 data source=<SQLServerName>;initial catalog=<InnovatorDatabase>
 e.g.
 data source=MyServer\MSSQLSERVER2019;initial catalog=InnovatorSolutions
4. Select the **Using the following credential** choice for the **Connect Using** field and enter the **DB Client Login** credentials for your database.
5. Click **Create** to deploy the shared Data Source.

3.4 Adding Reporting Services User and Role

The entry point for the Microsoft Reporting Services is the ReportServer.asmx file. The same Web Service is used to request Reports and administer the Report Server.

The Web Service is configured by default to use Windows Authentication. You can't simply open the security for anonymous Web users because that would open the Report Manager as well.

This security model requires the User be authenticated when they request Reports. A dialog opens requesting the user's login credentials the first time they request a Report each time the browser is opened again. When requesting the Report from within Aras Innovator this would not be desirable. Users would not only have to constantly provide login credentials to access reports, but each Innovator user would also have to have either a domain or local user access on the server.

To handle this problem, Aras provides the Innovator Reporting Services Gateway that processes requests to Reporting Services. This gateway uses one shared User Account for all Report requests. The Gateway submits the actual Report request using this shared User Account and propagates the results back to the client. The Report renders without the authentication dialog appearing.

If the user attempts to view the Report directly outside Aras Innovator, they still see the authentication dialog as normal. So, users can only view the Reports through Aras Innovator.

The local User Account on the server by default is named 'Reporting' with a default password is 'reporting' and must be mapped as a Role in the Report Server.

3.4.1 Add User Account for Reporting

The administrator for the server where Microsoft Reporting Services is installed must add a new Windows User Account named 'Reporting'. Once the User Account is created Aras Innovator needs to be configured to use account credentials.

3.4.1.1 Configure User Account for On-Premises Innovator instance

Reporting Service configuration is stored in InnovatorServerConfig.xml configuration file. <ReportingServices> section needs to be updated with the username and password:

```
<ReportingServices>
  <ReportServer>http://localhost/ReportServer/</ReportServer>
  <User>Reporting</User>
  <Password>reporting</Password>
  <Domain>Domain for the above User</Domain>
</ReportingServices>
```

Optional node, <Domain>, should contain the domain name you want to use to connect to Reporting services.

3.4.2 Add the Reporting Role

The **Reporting** User Account must next be mapped as a Role in Reporting Services enabling this account Browser privileges.

1. Create the **Reporting** user on the Server.

2. Click the **Site Settings** link in the upper right-hand corner of the **Report Manager** page.

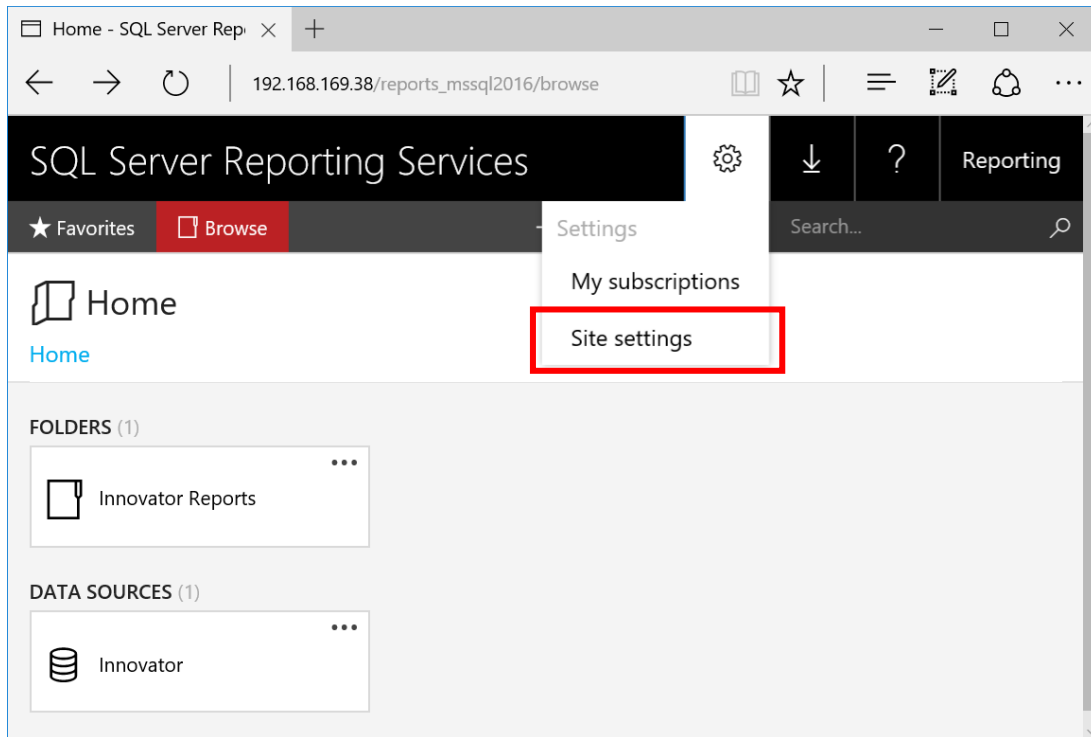


Figure 24.

3. Click the **Security** tab.

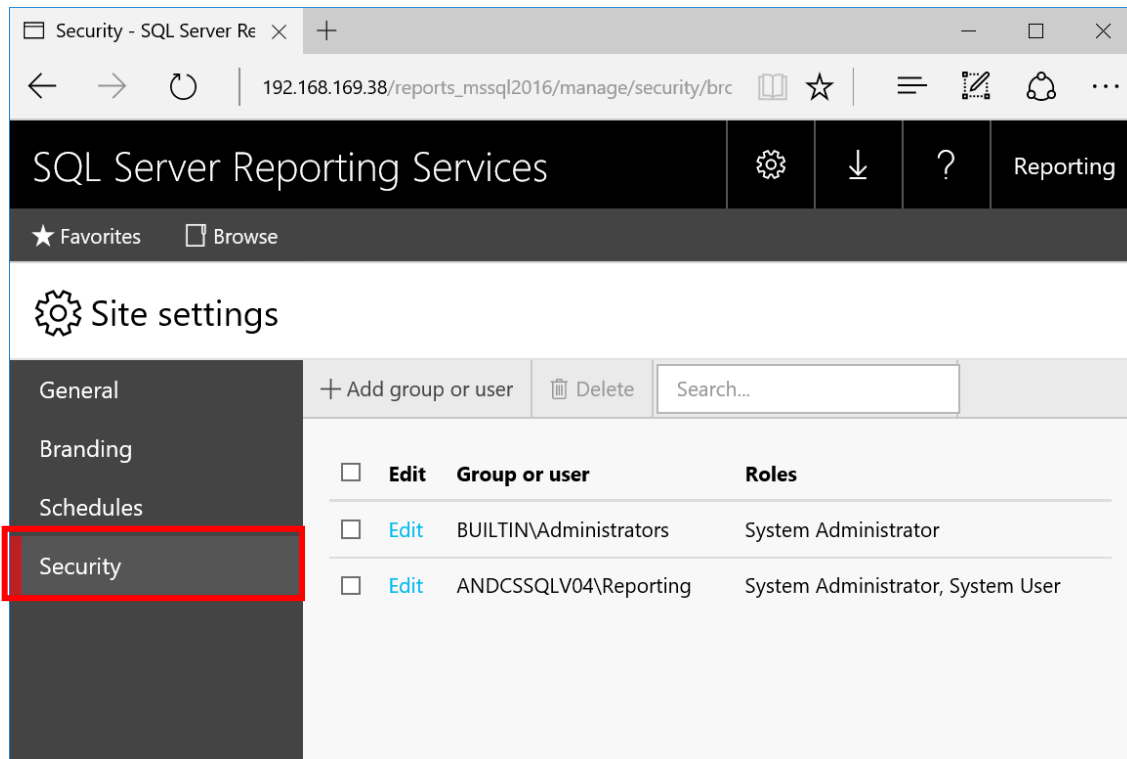


Figure 25.

4. Click **Add group or user** to map the **Reporting** User Account as a Reporting Services Role.

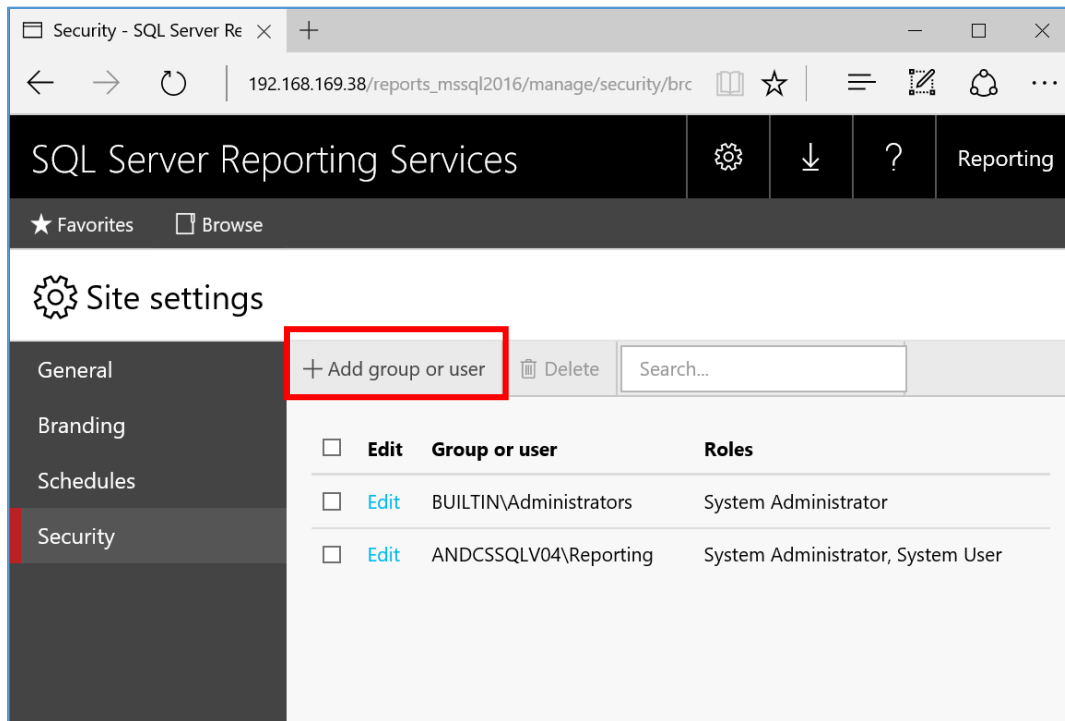


Figure 26.

5. Enter the **Reporting** User Account name, select the **System** checkbox, and click **OK** to create the new Role.

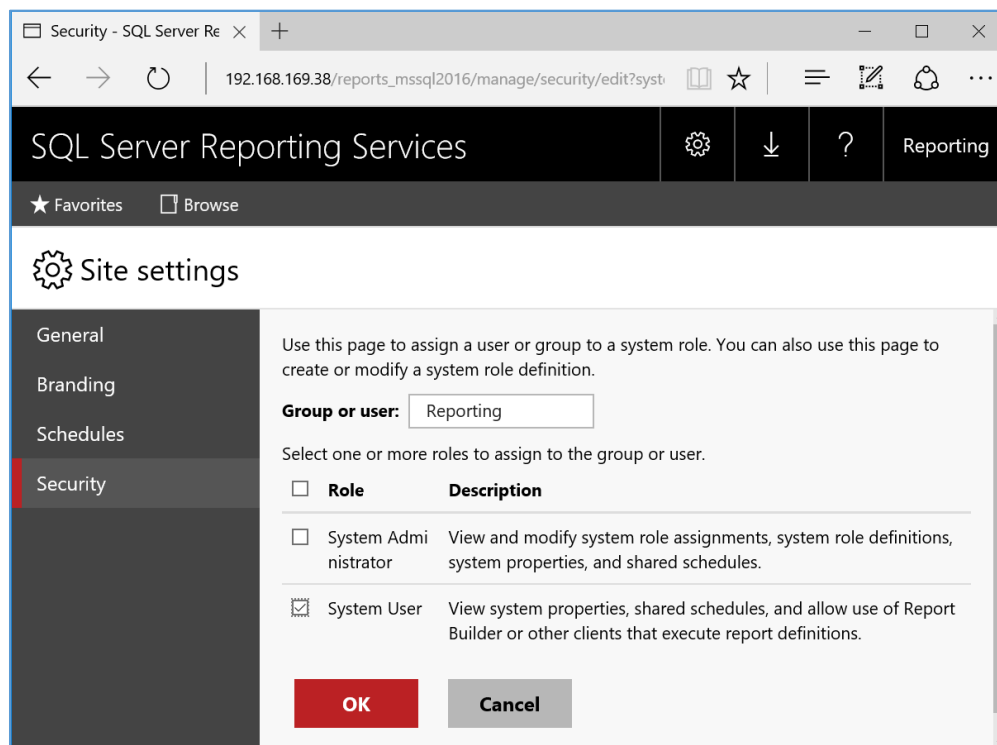


Figure 27.

6. Click the **Browse** link in the upper left corner of the page.

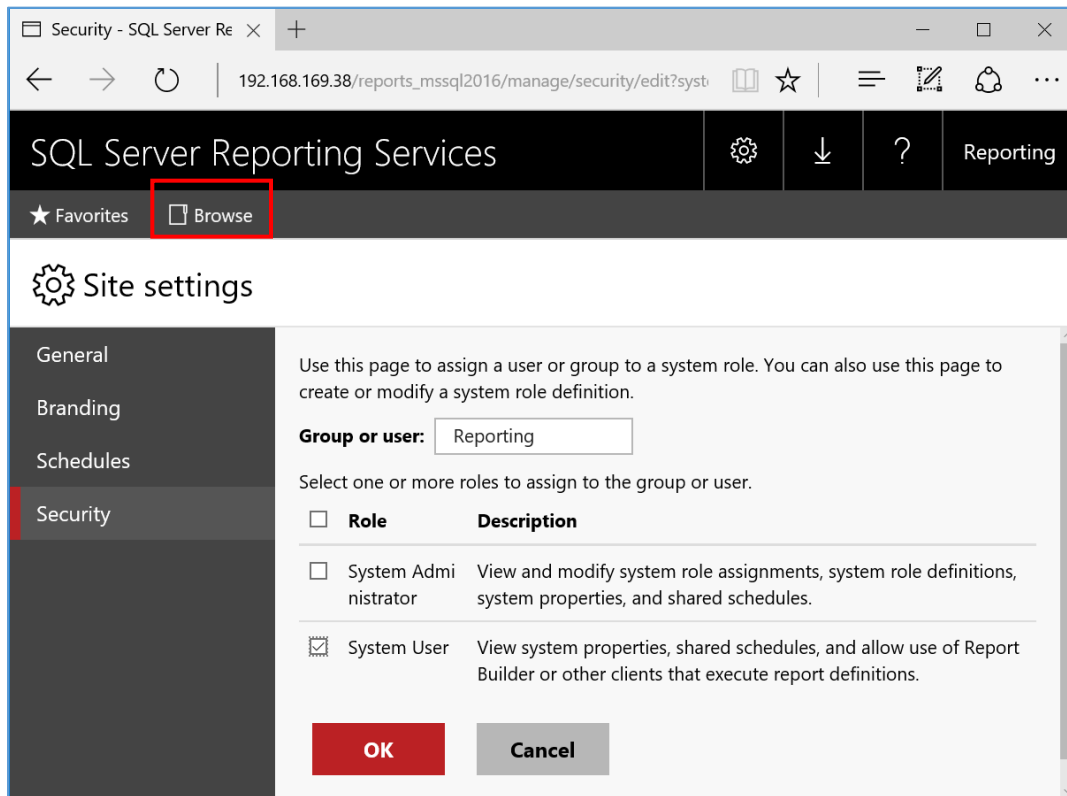


Figure 28.

7. Click the **Manage Folder** link.

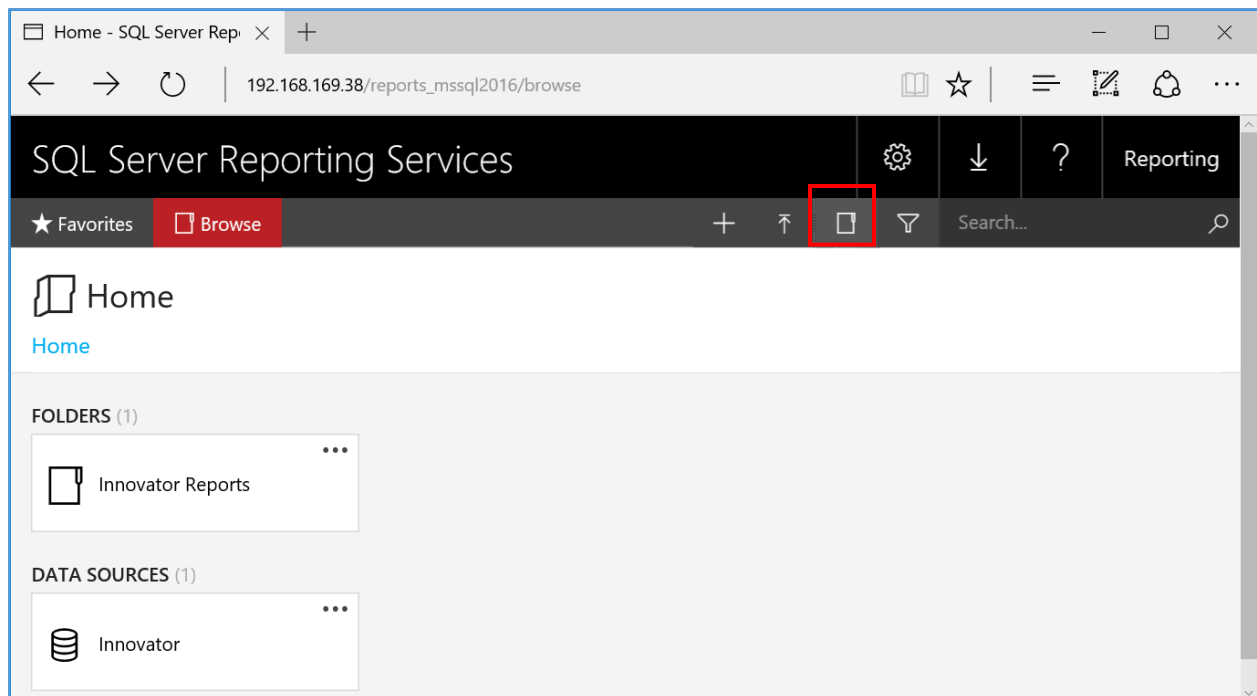


Figure 29.

8. In the settings tab, click **Add group or user**.

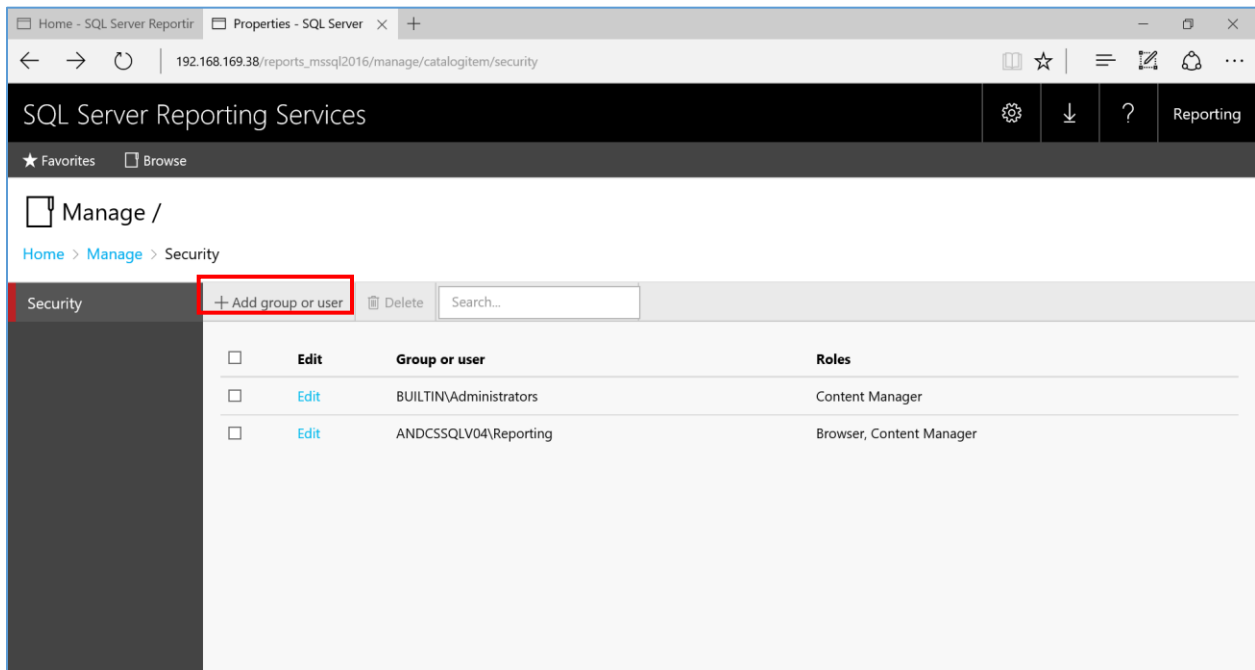


Figure 30.

9. Enter **Reporting** in the **Group or user** field, select the **Browser** checkbox, and click **OK** to create a new Role Assignment for the **Innovator Reports** folder.

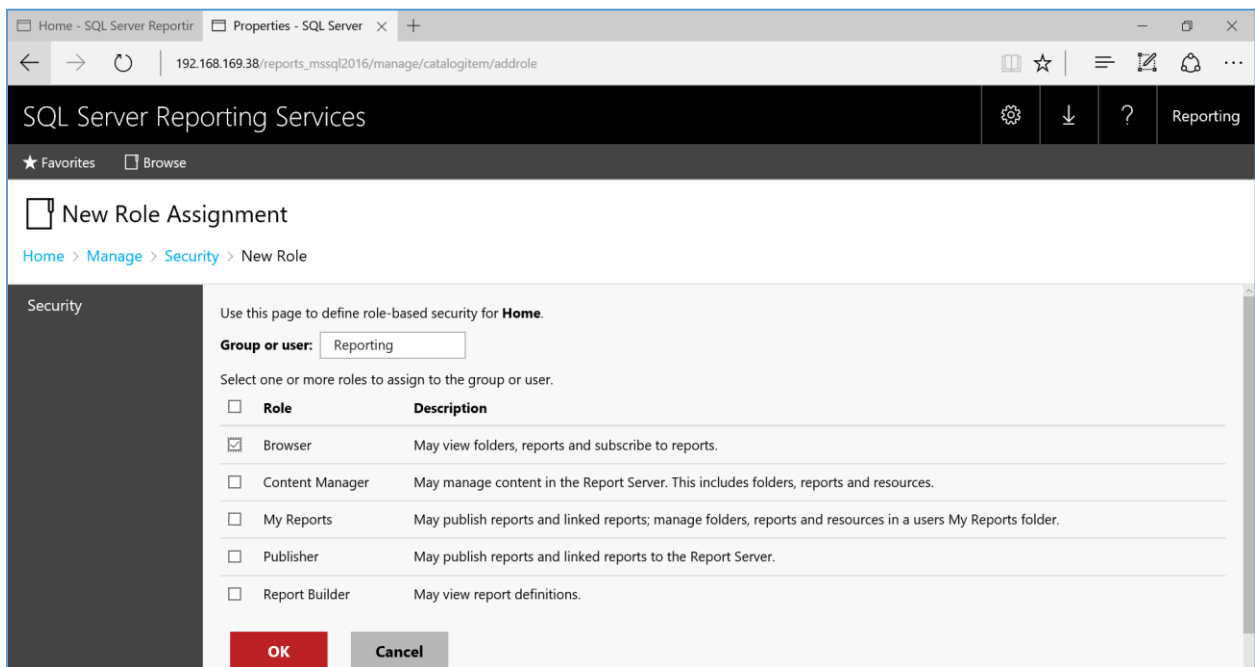


Figure 31.

4 Using Visual Studio for Report Design

Once the Reporting Services is installed, you need to configure Visual Studio to create a Project to hold the Reports. You also need to configure a Shared Data Source, which is the database connection information for the Reports.

Note: It is required that you follow the exact naming conventions as shown here to ensure that the Aras Innovator integration with the Reporting Services works correctly.

4.1 Create a New Report Project

Start **SQL Server Data Tools** and create a new **Innovator Reports** project.

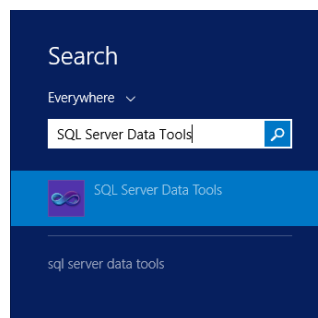


Figure 32.

This becomes the **Reporting Services** Folder to hold the Reports for Aras Innovator.

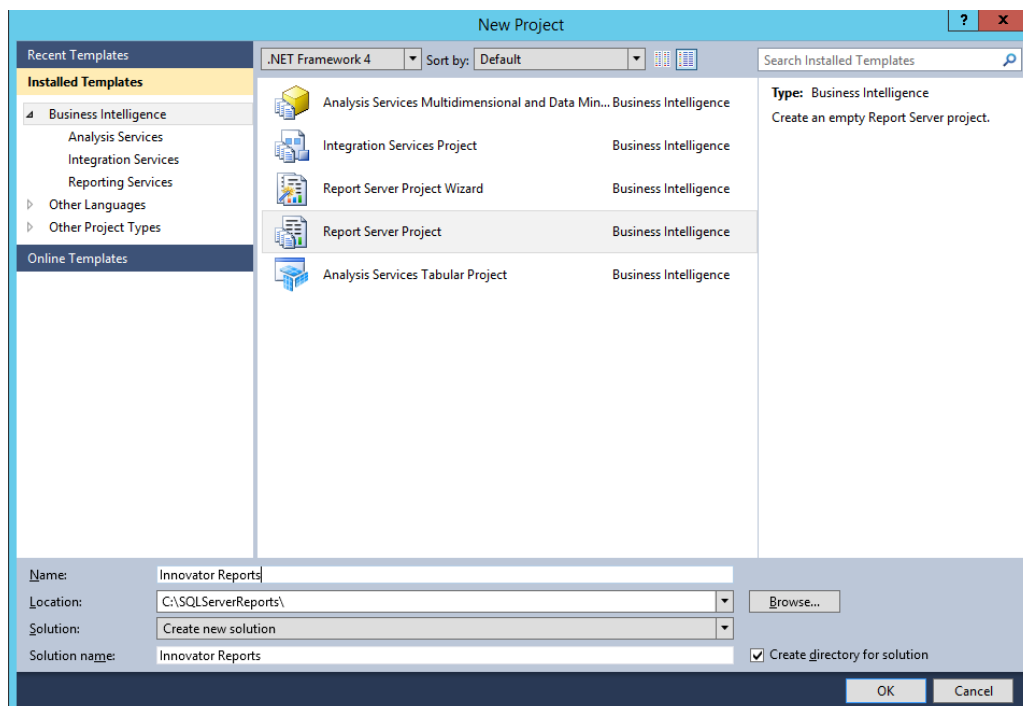


Figure 33.

4.2 Configure Project Properties

1. Open the **Solutions Explorer** panel from the **View --> Solutions Explorer** menu if it is not already opened.
2. Right-click the **Innovator Reports** folder and select the **Properties** context menu choice.
3. In the **Properties** dialog, set the **TargetServerURL** field to the URL of the Report Server. For example, <http://localhost:80/ReportServer>, where the actual name of the server host name is entered in place of localhost.
4. Set the **OverwriteDataSources** flag to **True**.

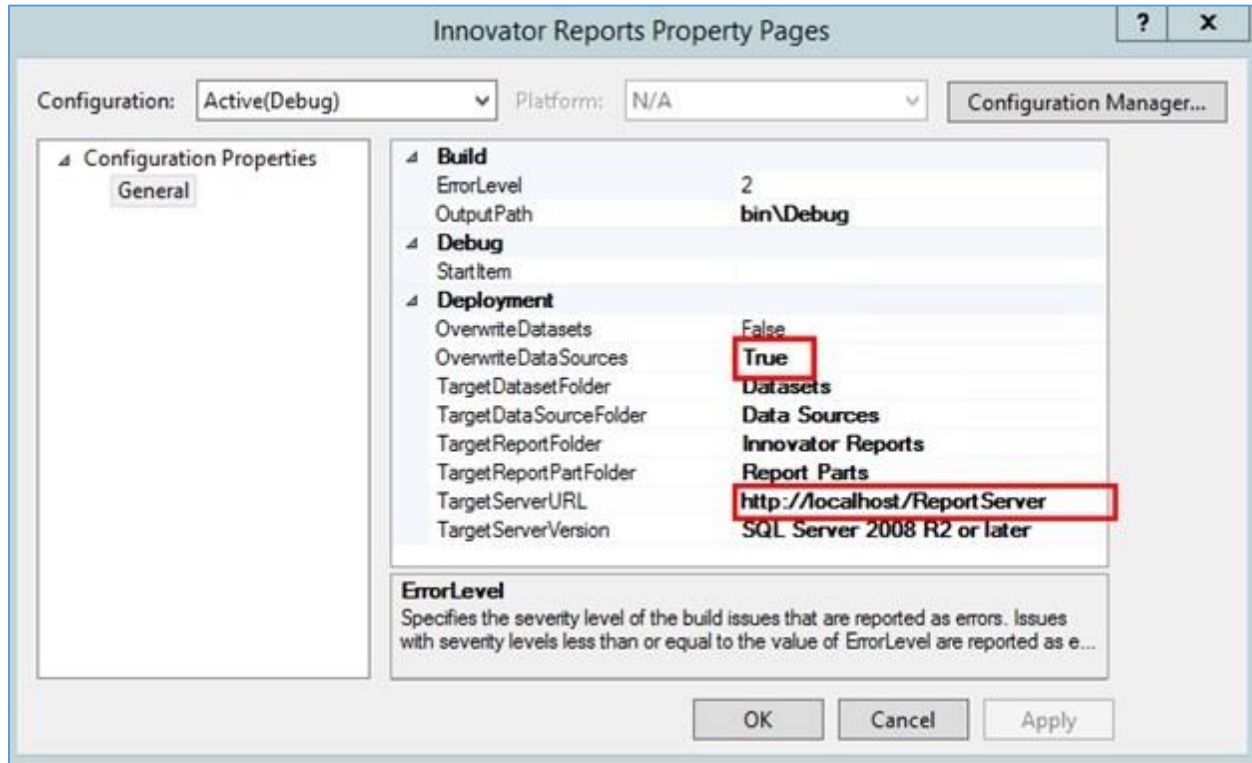


Figure 34.

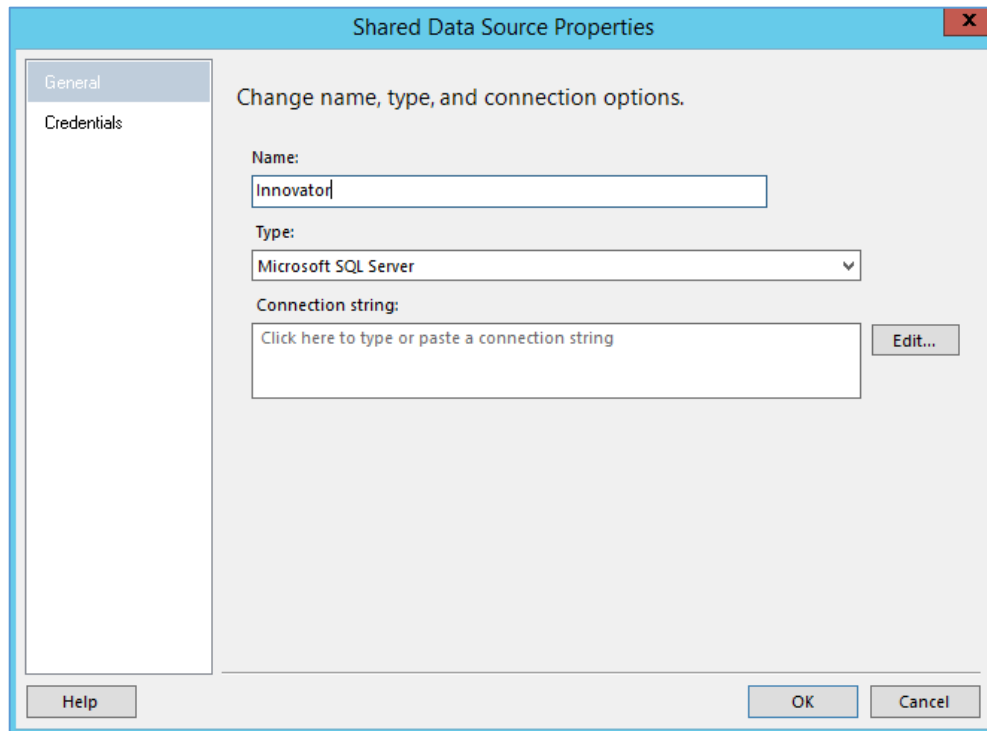
4.3 Add New Shared Data Source

In the same **Solutions Explorer** panel, right-click the **Shared Data Sources** folder, select **Add New Data Source** context menu choice, and click **Edit** to configure the new data source.

4.3.1 Connection Properties

1. Select or enter a server name for the machine with the installed SQL Server.
2. Select the **Use a specific user name and password** option, enter login credentials for the database, and select the **Save My password** option.

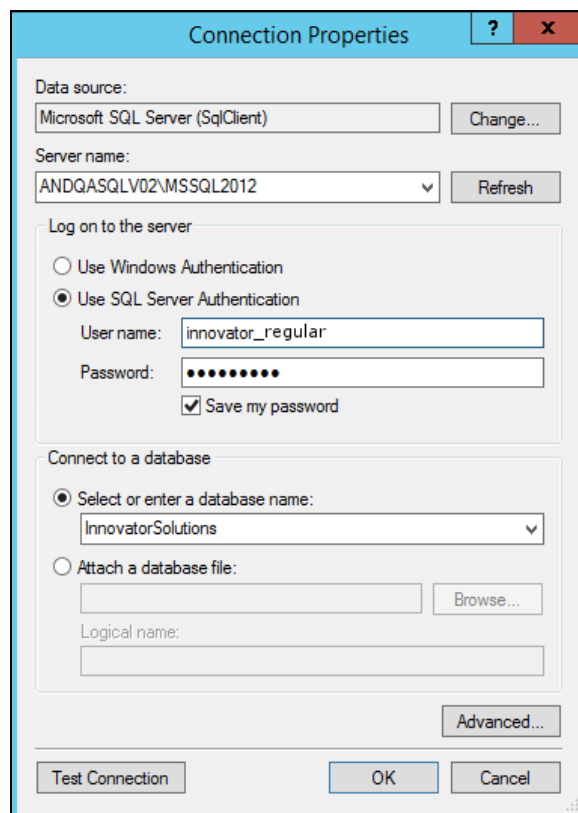
3. Select or enter a database name.



The 'Shared Data Source Properties' dialog box is shown with the 'General' tab selected. It contains the following fields and controls:

- Name:** A text box containing 'Innovator|'.
- Type:** A dropdown menu set to 'Microsoft SQL Server'.
- Connection string:** A text box with the placeholder 'Click here to type or paste a connection string' and an 'Edit...' button to its right.
- Buttons:** 'Help', 'OK', and 'Cancel' are located at the bottom.

Figure 35.



The 'Connection Properties' dialog box is shown with the following fields and controls:

- Data source:** A dropdown menu set to 'Microsoft SQL Server (SqlClient)' with a 'Change...' button to its right.
- Server name:** A dropdown menu set to 'ANDQASQLV02\MSSQL2012' with a 'Refresh' button to its right.
- Log on to the server:** A section with two radio buttons: 'Use Windows Authentication' (unselected) and 'Use SQL Server Authentication' (selected). Below these are text boxes for 'User name:' (containing 'innovator_regular') and 'Password:' (masked with dots), and a checked checkbox for 'Save my password'.
- Connect to a database:** A section with two radio buttons: 'Select or enter a database name:' (selected) and 'Attach a database file:' (unselected). The 'Select or enter a database name:' option has a dropdown menu set to 'InnovatorSolutions'. The 'Attach a database file:' option has a 'Browse...' button and a 'Logical name:' text box.
- Buttons:** 'Test Connection', 'OK', and 'Cancel' are at the bottom. An 'Advanced...' button is located above the 'OK' and 'Cancel' buttons.

Figure 36.

4.3.2 Deploy the Shared Data Source

By default, the name of the Shared Data Source is the name of the Database you selected previously. You can rename this if you want. Finally, you need to deploy this new Shared Data Source.

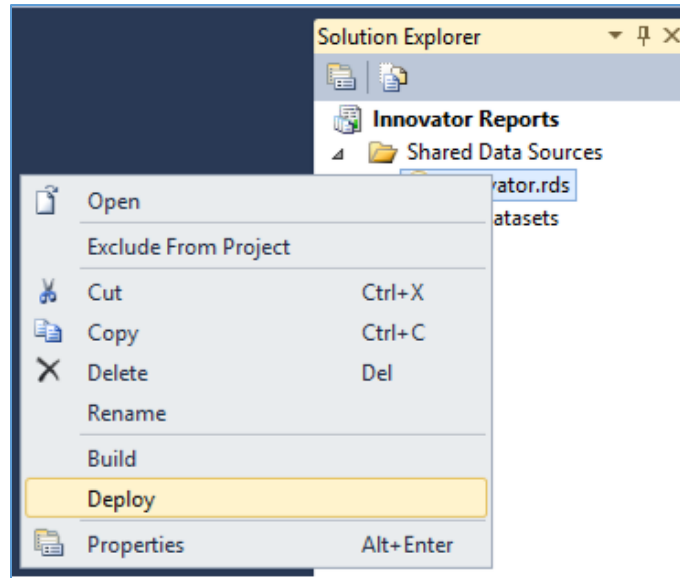


Figure 37.

5 Creating your First Report

This section describes how to create a new Report using the Report Wizard. We will create a simple User Directory Report showing the User's First Name, Last Name, Email, Phone Number, and Company Name.

5.1 Start a New Report using the Wizard

In the **Solution Explorer** panel, right-click the **Reports** folder, and select the **Add New Report** context menu choice.

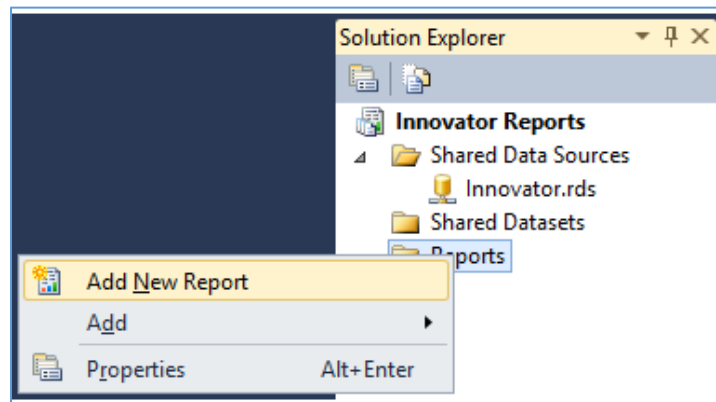
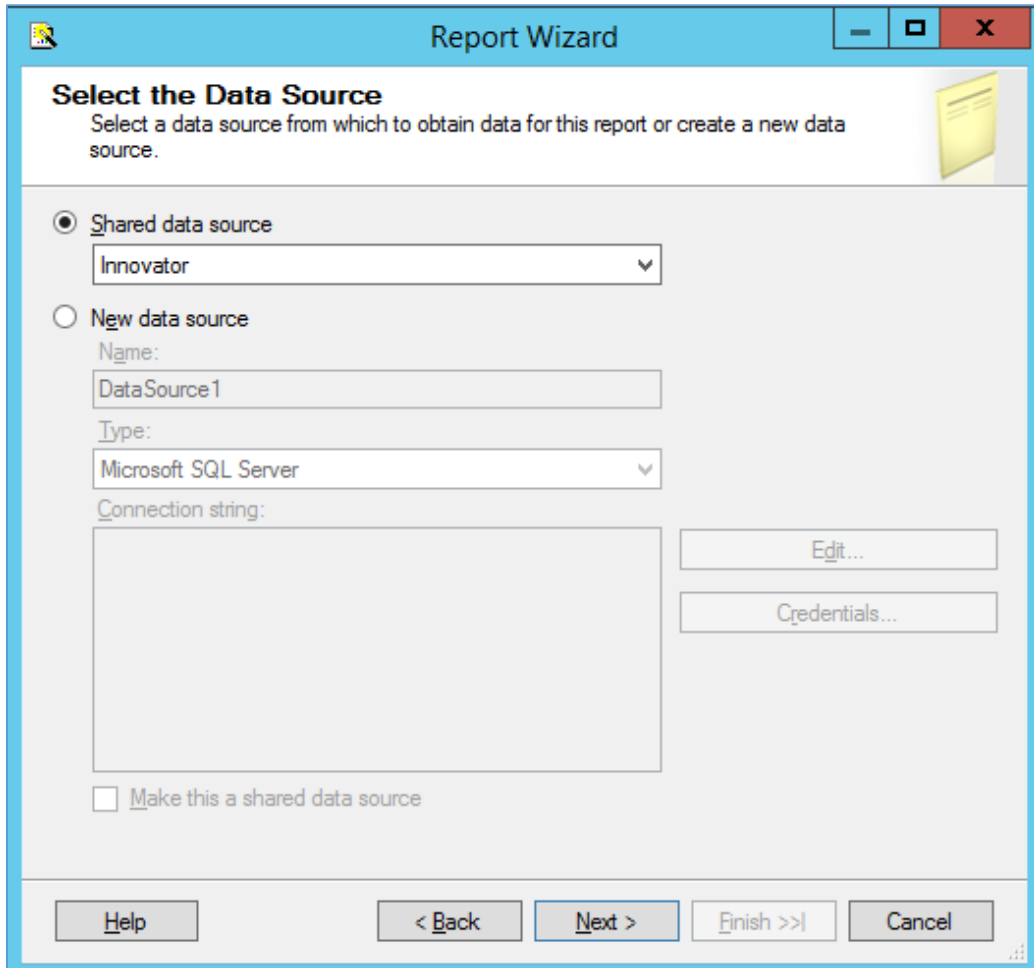


Figure 38.

5.2 Select the Data Source

You can create a unique Data Source for this Report, but you would typically use the default Shared Data Source that you created.



The screenshot shows the 'Report Wizard' window with the 'Select the Data Source' step. The window has a blue title bar with standard Windows controls. The main area is light gray with a blue border. At the top, the title 'Select the Data Source' is in bold, followed by the instruction 'Select a data source from which to obtain data for this report or create a new data source.' Below this, there are two radio buttons: 'Shared data source' (selected) and 'New data source'. Under 'Shared data source', there is a dropdown menu showing 'Innovator'. Under 'New data source', there are fields for 'Name:' (containing 'DataSource1'), 'Type:' (a dropdown menu showing 'Microsoft SQL Server'), and 'Connection string:' (a large text area). To the right of the 'Connection string' field are two buttons: 'Edit...' and 'Credentials...'. At the bottom left of the main area is a checkbox labeled 'Make this a shared data source'. At the bottom of the window is a navigation bar with five buttons: 'Help', '< Back', 'Next >', 'Finish >>', and 'Cancel'.

Figure 39.

5.3 Design the Query

You need to provide the `SELECT` SQL query that fetches data for the Report. In many cases, you can enter a simple `SELECT` clause directly in the dialog box.

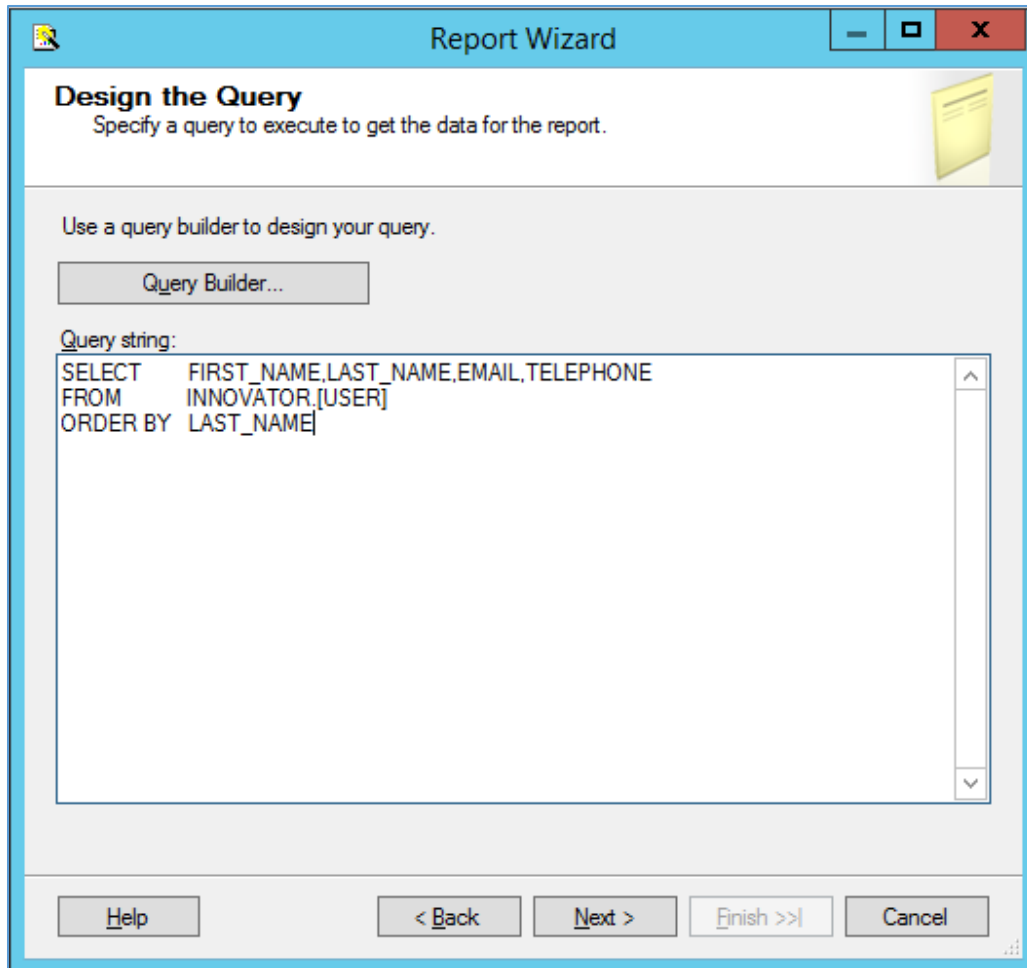


Figure 40.

Note: It is possible to pass Named Parameters into the query such as:

```
SELECT * FROM [PROJECT] WHERE id=@ID
```

The **Edit...** button opens a Query Builder interface, which is covered later in more detail.

5.4 Select the Report Type

There are two choices for the Report Type:

- **Tabular**: column layout.
- **Matrix**: rows and columns layout.

Select the **Tabular** type.

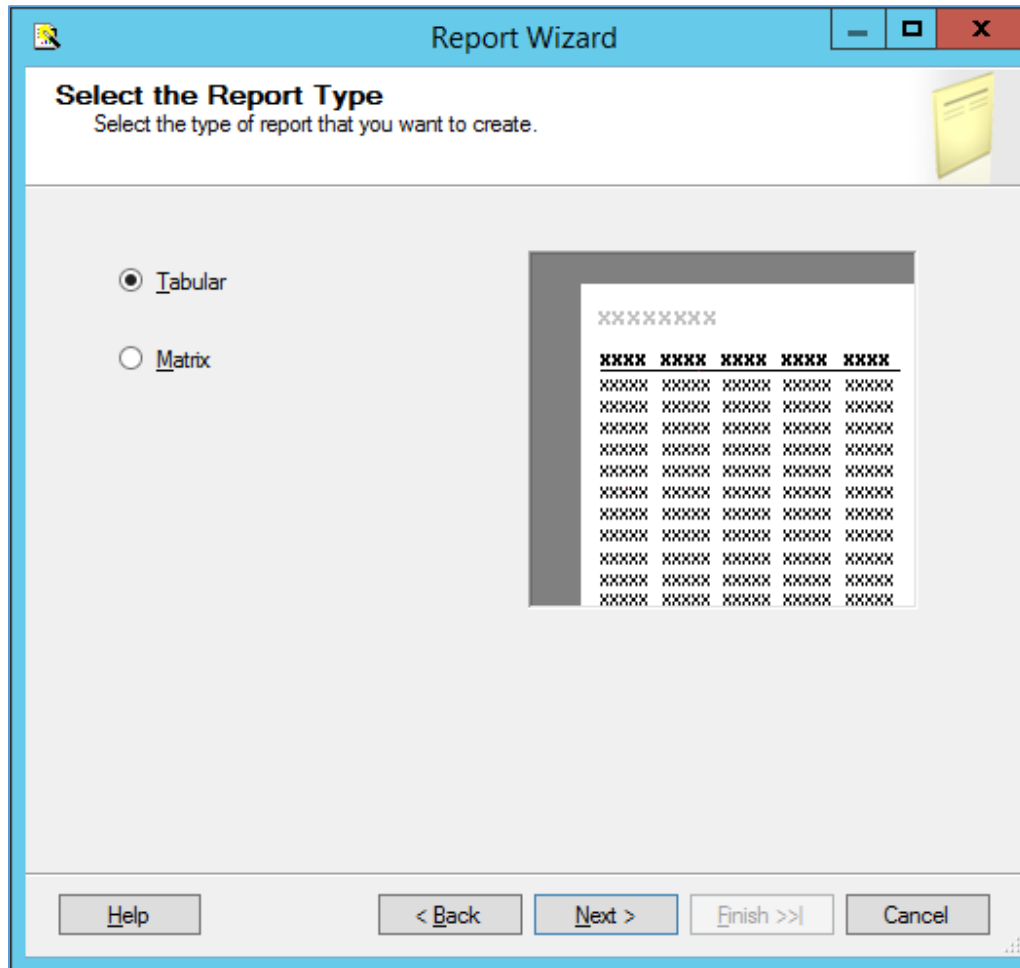


Figure 41.

5.5 Design the Table

Select the fields that you want to show on the Report. In this case, we were selective about which fields to return from the query select clause, so we want all the fields.

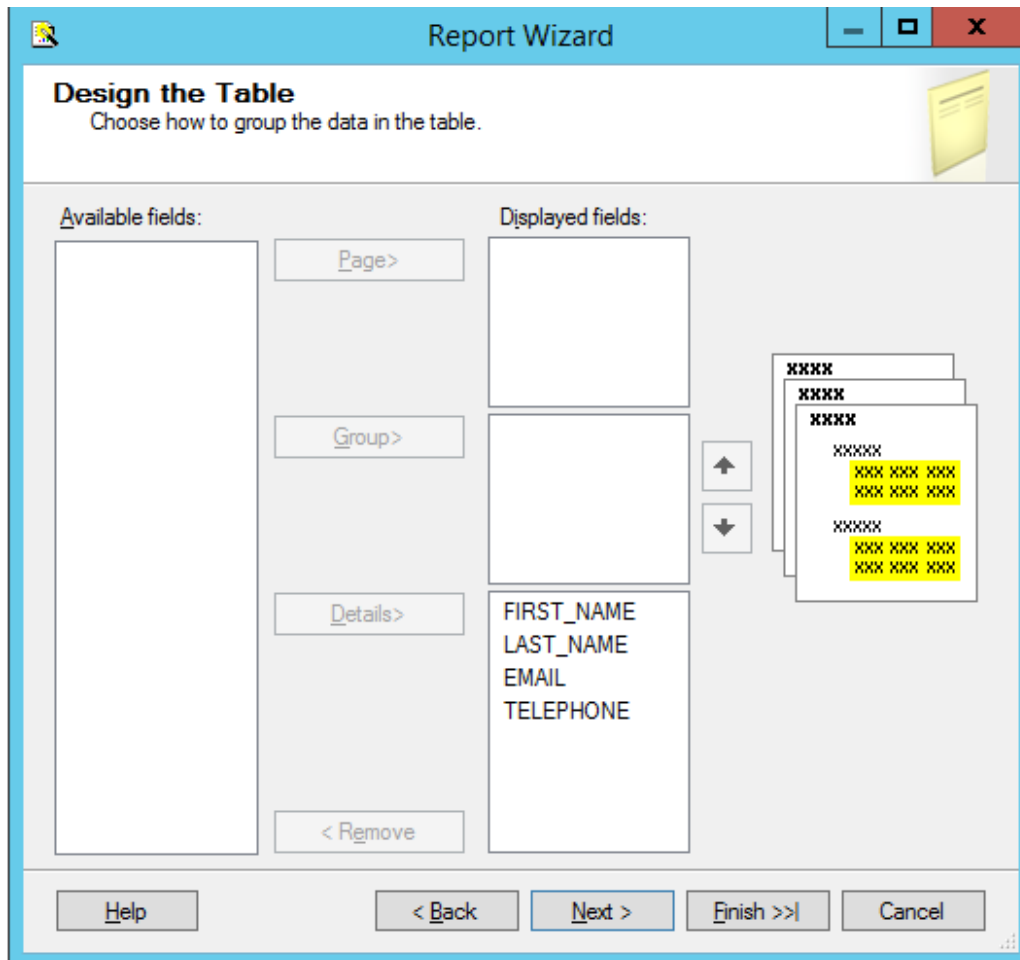


Figure 42.

5.6 Choose the Table Style

Select the table style for the Report. This example simply uses the default **State** style.

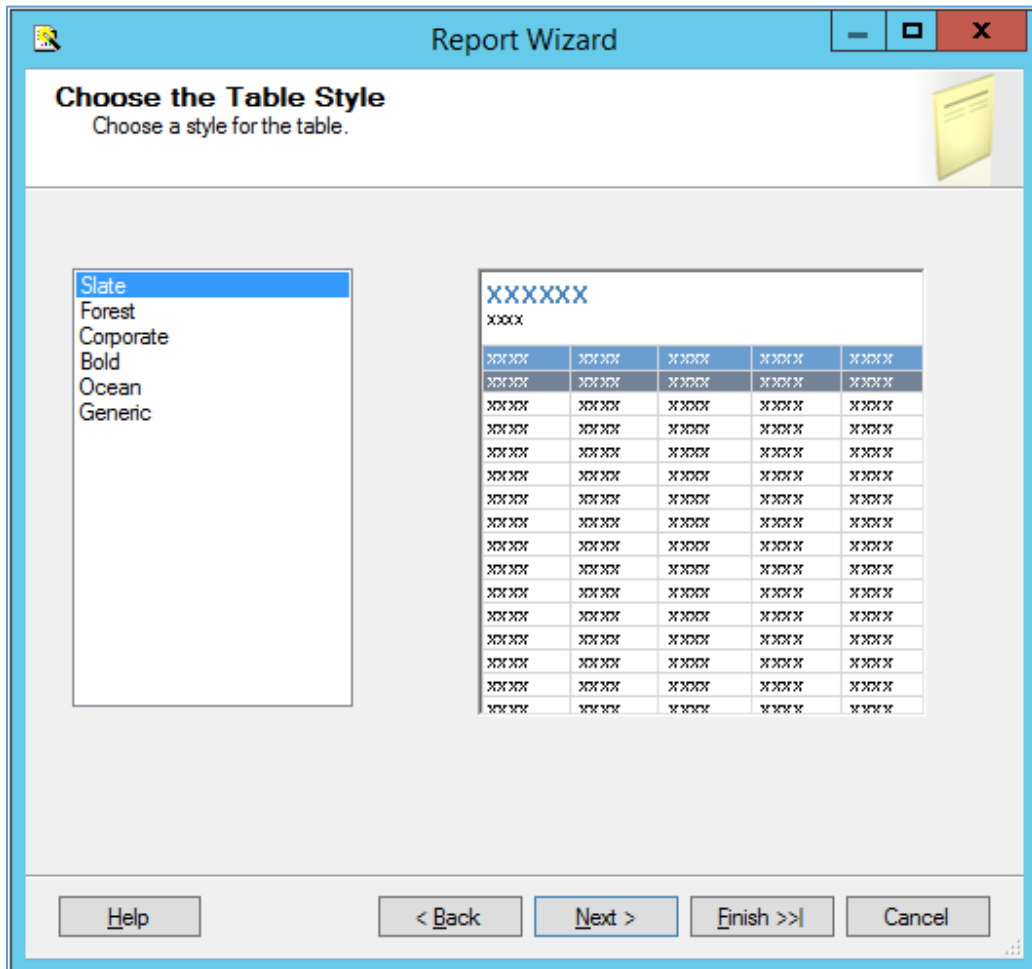


Figure 43.

5.7 Name the Report

The final step in the Report wizard is to name the Report.

Note: The chosen **Report Name** must be meaningful and should be the same name you set as the name of the **Report Item** in Aras Innovator.

Report Wizard

Completing the Wizard
Provide a name and click Finish to create the new report.

Report name:
User Report

Report summary:

Data source: Innovator
Connection string:
Report type: Table
Layout type: Stepped
Style: Slate
Details: FIRST_NAME, LAST_NAME, EMAIL, TELEPHONE
Query: SELECT FIRST_NAME, LAST_NAME, EMAIL, TELEPHONE
FROM INNOVATOR.[USER]
ORDER BY LAST_NAME

☒ Preview report

Help < Back Next > Finish Cancel

Figure 44.

5.8 Wizard Report Results

The Report wizard builds the report, which looks like Figure 45.

User Report			
FIRST NAME	LAST NAME	EMAIL	TELEPHONE
ES	Admin		
Innovator	Admin		
Vault	Admin		
Super	User		

Figure 45.

Once the Report is opened in the **Layout** Tab, you can make further changes as required.

5.9 Visual Studio Layout

The **Report** pane has three Tabs:

- **Data:** for editing the query for the Report. Look at the appendix in this document to see a more complex Report query example, using the graphical query builder instead of the wizard.
- **Layout:** for visually designing the Report with drag and drop functionality.
- **Preview:** for viewing the finished report with real data.

The **Properties** panel on the right is used for setting fonts, color, and other visual properties.

The **Fields** panel on the left is used to drag database fields onto the report.

The **Toolbox** panel also on the left is for dragging Report controls onto the **Report** layout. This is how you can add tables, labels, images, and charts to the **Report** layout.

Note: Use **View --> Fields** to open the **Fields** panel when in the **Layout** mode.
Use **View --> Toolbox** to open the **Toolbox** panel.
To open the Header and Footer, right-click to open the context menu in the pink field in the **Report** layout.

5.10 Deploying the Report

Once you are satisfied with the Report query and layout as viewed from the **Preview** Tab, you need to deploy the Report to the Reporting Service:

- In the **Solution Explorer** panel, right-click the User Directory Report item and select the **Deploy** context menu choice.

Watch for error messages in the output window at the bottom of the frame.

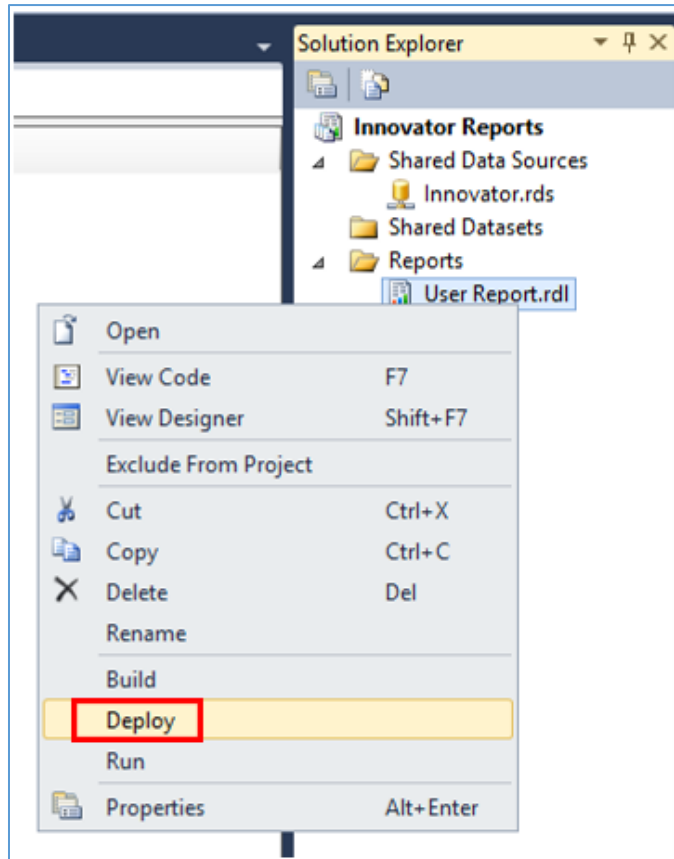


Figure 46.

5.11 Confirm Deployment

To confirm the new Report has been deployed:

1. Open a browser and go to <http://localhost:80/Reports>.
You should see the **Innovator Reports** folder.
2. Click that link.
You should see the report you created.
3. Select the hyperlink for the new Report.
4. Run the Report.

5.12 Running the Report from within Aras Innovator

For users to run the new Reporting Services Report from within Aras Innovator, you need to create an **Innovator Report** Item. The important settings to enable the Innovator Reporting Services interface are:

- The **Innovator Report** Item name must be the same as the Reporting Services Report name.
- The **Location** Field choice must be set to the new **Service** option.
- The **Report Query** Field specifies the parameters that are passed to the Report Server.

5.12.1 Adding Aras Innovator Report Item

1. Log into Aras Innovator as administrator.
2. In the **Navigation** pane, go to **Administration --> Reports**. The menu shown in Figure 47 appears.

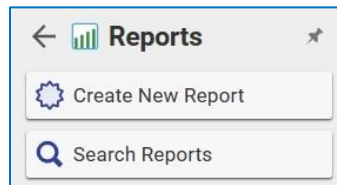


Figure 47.

3. Click **Create New Report**. A blank Report form appears:

Figure 48.

4. Enter the name of the Report.

Note: Again, this is the name you specified when you created the Report in Visual Studio.

5. Select the Report Type, which in this case is **Generic**, meaning this Report is always available.
6. Select the Report Location, which in this case is **Service**.
7. Select the Report Target, which in this case is **Window**.

Because this sample Report is of type ItemType, there is no Report query Value.

Note: If you use Firefox, you have to manually enter a stylesheet under the **Stylesheet** tab.

8. Test the Report by clicking the **Reports** main menu bar choice.
 You should see the new **User Report** choice.
9. Click the **menu** choice.
 You should see a new dialog opened with the **User Directory** Report.

5.12.2 Passing Named Parameters to the ReportServer

As step 4 describes in section 3.1 The Reporting Services use Folders to organize Reports. The Aras Innovator integration with Reporting Services use as the default **Innovator Reports** Folder.

In this section, we configure the Reporting Services using the Report Manager by adding the **Innovator Reports** Folder and a Shared Data Source, which the Reports use to connect to the database.

Installing Reporting Services, the value of the **report_query** Innovator Report Item Property is appended to the ReportServer URL, providing additional parameters to control the ReportServer.

For example, the ID of the Item can be passed as the Named Parameter @ID for the SQL Select for the Reporting Services Report as in this example:

- `SELECT * FROM [PROJECT] WHERE id=@ID`

To pass the ID for the context Item (the selected Item) on the URL to the ReportServer, the Innovator Report Item **report_query** Property would look like this:

- `np:id=<xsl:value-of select="@id"/>`

This is actually an XSLT stylesheet fragment and is the directive for Innovator to add the string `&id=ACBDEF0123456789ACBDEF0123456789` to the end of the ReportServer URL (but the actual ID value is used).

In this case, the Innovator Report is of the **Generic** type, so no ID is required, but perhaps we want the Reporting Services Report to be returned in the PDF or Excel format. We can control the ReportServer by adding the following parameter as the **report_query** value for the **User Directory** Report:

- `rs:Format=PDF`

Figure 49.

Note: If you need to include more than one parameter for the **report_query** value, you need to use the **&#amp;** Entity rather than the actual **&** symbol to delimit the parameters. Recall that the **report_query** value is eventually part of the **query_string** for the URL to the ReportServer but is actually an XSLT stylesheet fragment and so the special symbols like **<**, **>**, and **&** need to be protected in XML need to be entered as Entities **<**, **>**, and **&#amp;**, respectively.

For example, to pass the Named Parameter ID and the Format PDF in the same **report_query**, the value would look like this:

- `np:id=<xsl:value-of select="@id"/>&#amp;rs:Format=PDF`

6 Report Server query_string Parameters

There are several parameters that you can use to control how the Reporting Services run your Report requests. You include these in the **report_query** Innovator Report Item property, which the client then uses to form **query_string** for the **ReportServer** request.

6.1 Named Parameters

Named Parameters can be passed into the **SELECT** SQL query. For example:

- `np:last_name=Kohl`

They would map the **Kohl** value to the **@last_name** Named Parameter in the **SELECT** SQL:

- `SELECT * FROM [USER] WHERE last_name=@last_name`

6.2 Format Parameter

Use the `Format` parameter to control the format the Report is rendered:

- `rs:Format=XML`
- `rs:Format=PDF`
- `rs:Format=HTML4.0`
- `rs:Format=MHTML`
- `rs:Format=EXCEL`
- `rs:Format=Word`

6.3 Zoom

Use the `Zoom` parameter to control the zoom:

- `rc:Zoom=100`
- `rc:Zoom=% zoom as integer`
- `rc:Zoom=Page Width`
- `rc:Zoom=Whole Page`

6.4 Reports Folder

The default Folder where Reporting Services Reports are deployed is in **Innovator Reports**; however, you may deploy the Reports in other Folders, and you specify this using this Parameter:

- `irs:Folder=Your Folder Name`

6.5 Report Name

The Aras Innovator client automatically appends the Parameter to specify the Report to run on **query_string**, which is the name of the **Innovator Report** Item. However, if you run reports programmatically, you may need to set this directly yourself:

- `irs:Report=Your Report Name`

7 Report Manager

The **ReportServer** Reporting Services comes with the **Report Manager** page to administer the Reports, Folders, Data Sources, and so on. This page has been integrated into the Aras Innovator client via an Innovator Report that opens the **Report Manager** page; see section 2 for more information about this **Report** Item.

Reports that are created and tested on the development machine can be moved to a production **ReportServer** by exporting an RDL file for a Report from your local development machine and then importing it into the production machine.

The process of exporting a Report, importing it into another **ReportServer**, and enabling the Report in Aras Innovator has four steps:

1. Export an RDL file via the **Report Manager** page.
2. Import the RDL file via the **Report Manager** page.
3. Reassign the Data Source to the imported Report.
4. Add an **Innovator Report** Item to call the **Reporting Services** Report.

7.1 Opening the Report Manager page

1. In the TOC, go to **Administration --> Reports**.
2. In the main menu, click **Reports** and the **Report Manager** choice.

The **Report Manager** page opens in a new dialog shown in Figure 41.

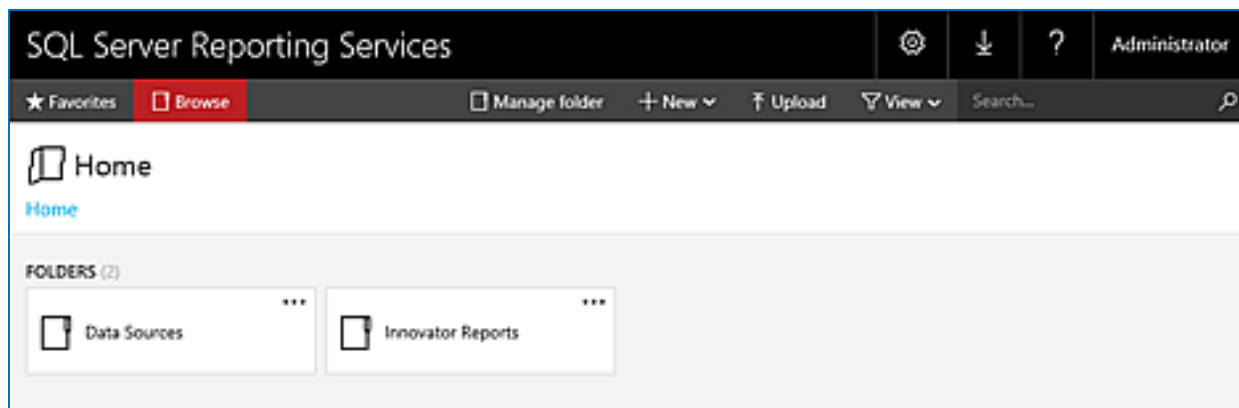


Figure 50.

Note: This page can be opened from your browser with <http://localhost:80/Reports>

7.2 Export the RDL

To export an RDL file for a Report:

1. Click the **Innovator Reports** folder.
You see a list of Reports and Shared Data Sources.
2. In the **Report** context menu, click **Manage**.

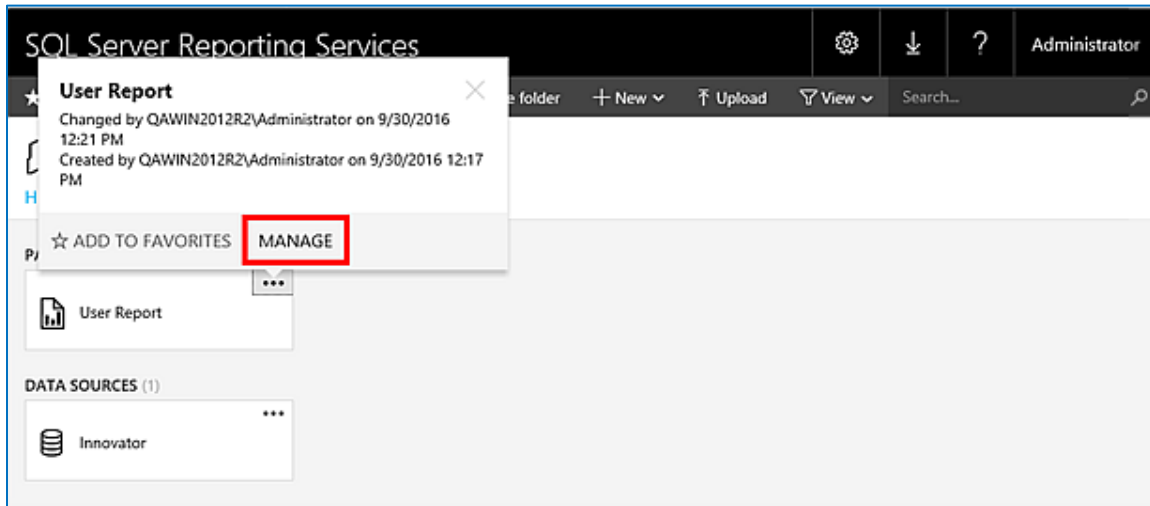


Figure 51.

3. In the **User Report** page, click **Download**.
This opens a dialog asking for confirmation that you want to save the RDL file.

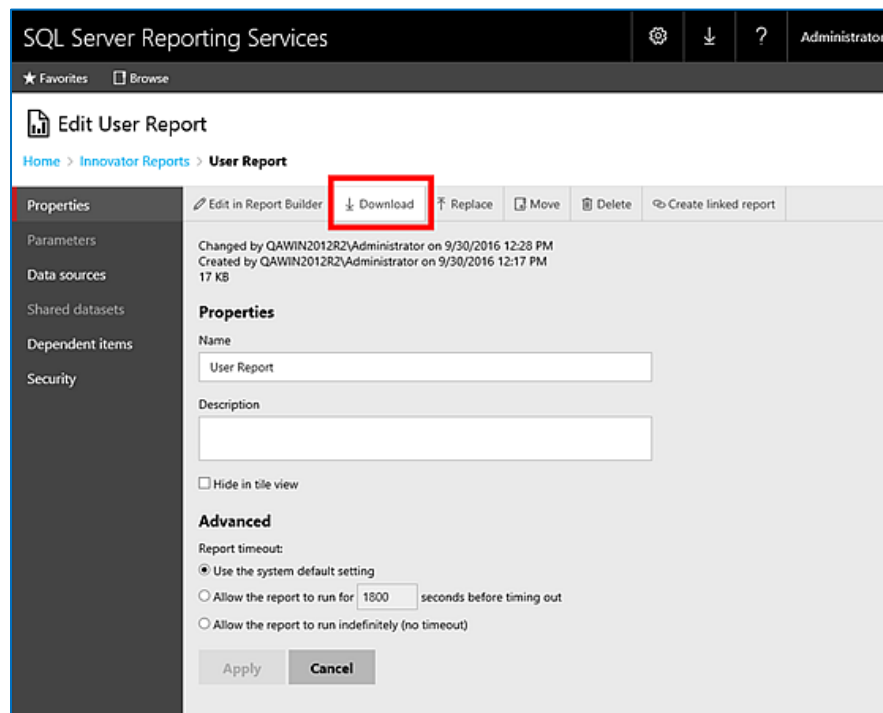


Figure 52.

7.3 Import the RDL

1. Log into the target Aras Innovator Server.
2. In the TOC, go to **Administration** --> **Reports**.
3. In the main menu, click **Reports** and then **Report Manager**.

The **Report Manager** page opens in a new dialog as shown in Figure 53.

Note: This page can be opened in your browser with <http://localhost:80/Reports>.

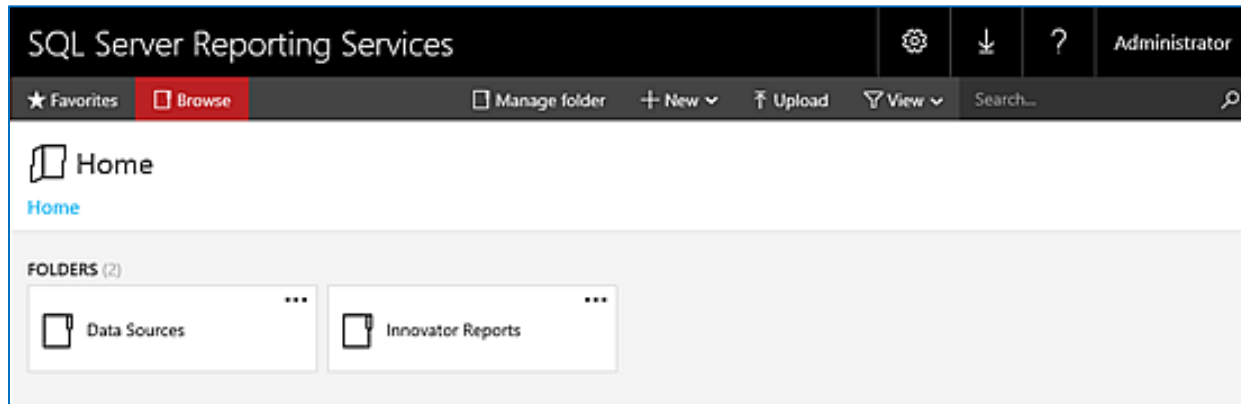


Figure 53.

4. Click the **Innovator Reports** Folder.
You see a list of Reports and Shared Data Sources.
5. Click **Upload** in the dialog middle.

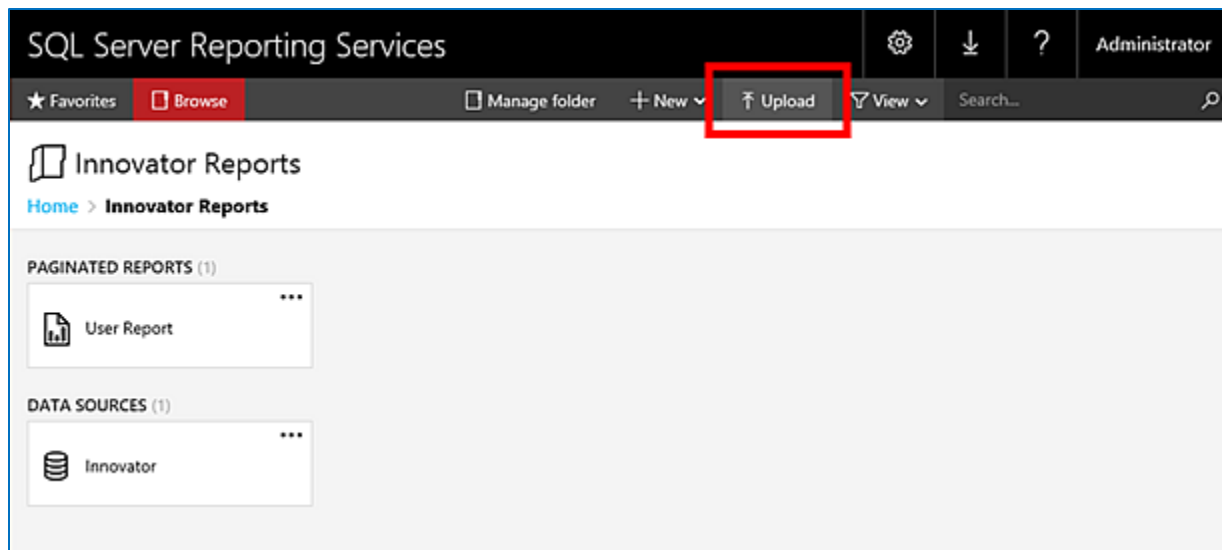


Figure 54.

This opens the **File Selection** dialog for you to select an RDL file to upload.

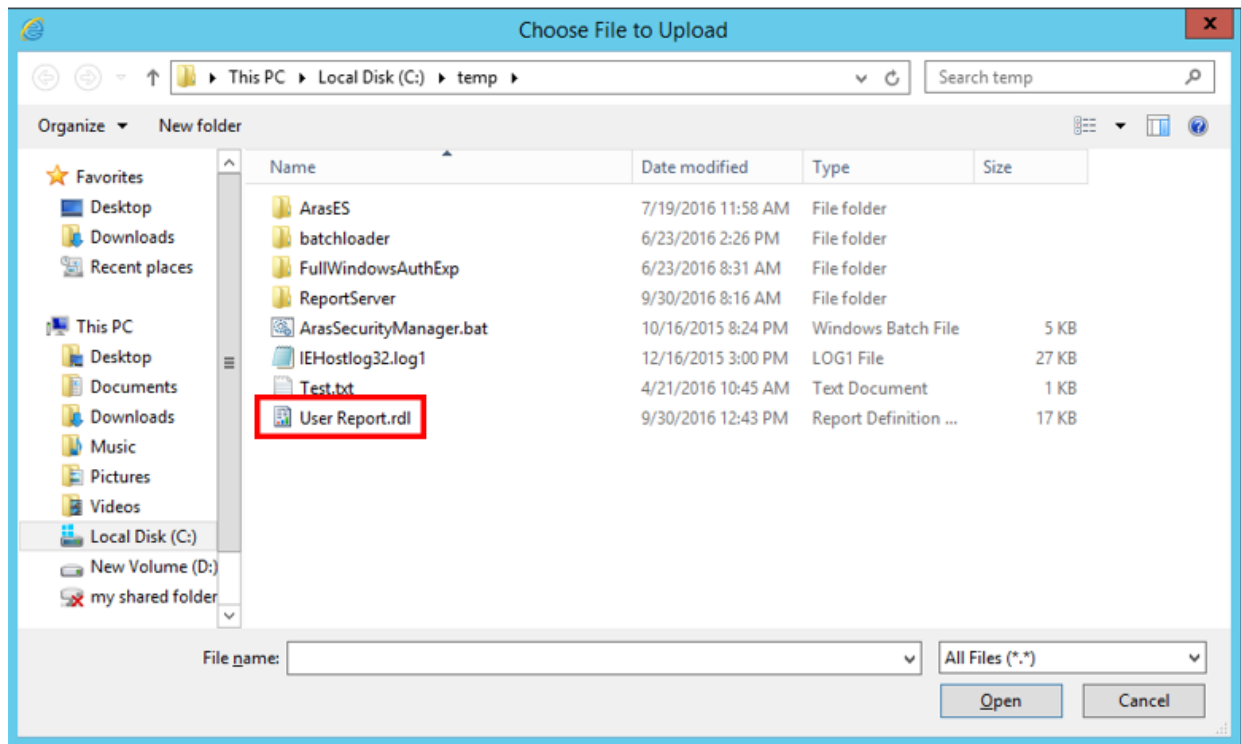


Figure 55.

6. Select the RDL file and click **Open**.

The RDL file is uploaded to the target ReportServer and deployed. You should see the new Report in the Folder now.

7.4 Select the Shared Data Source

The Data Source for the Report may need to be changed if the name of the Shared Data Source is different on the target ReportServer.

1. In the **Report** context menu, click **Manage**.

The **Properties** page for the Report appears where you can select its Shared Data Source.

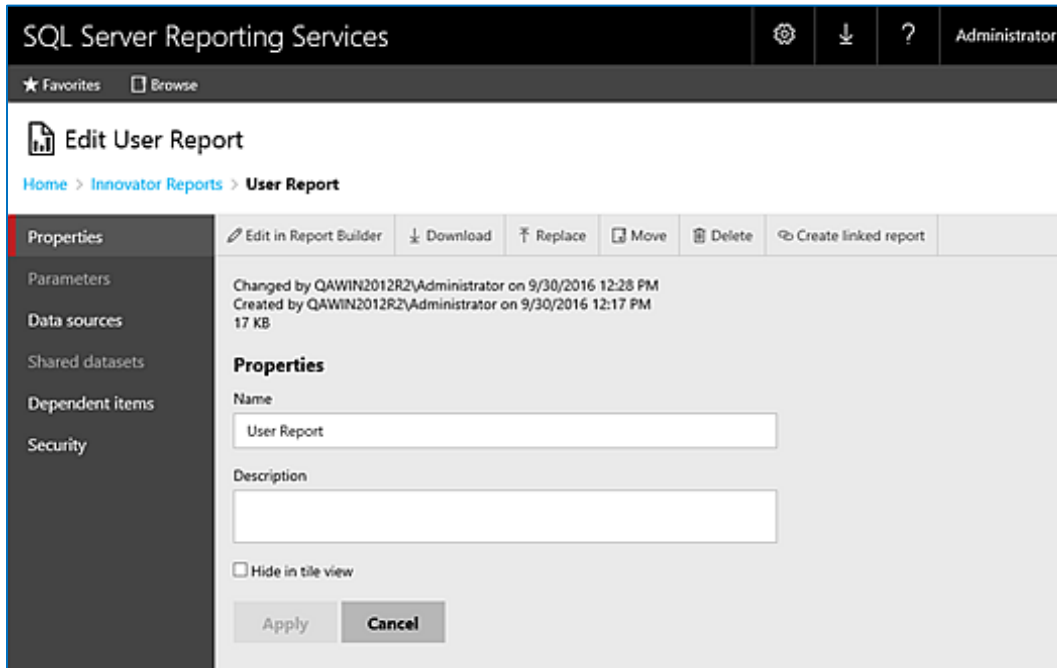


Figure 56.

2. Click the **Data Sources** navigation link.
3. Select the first **A shared data source** choice.

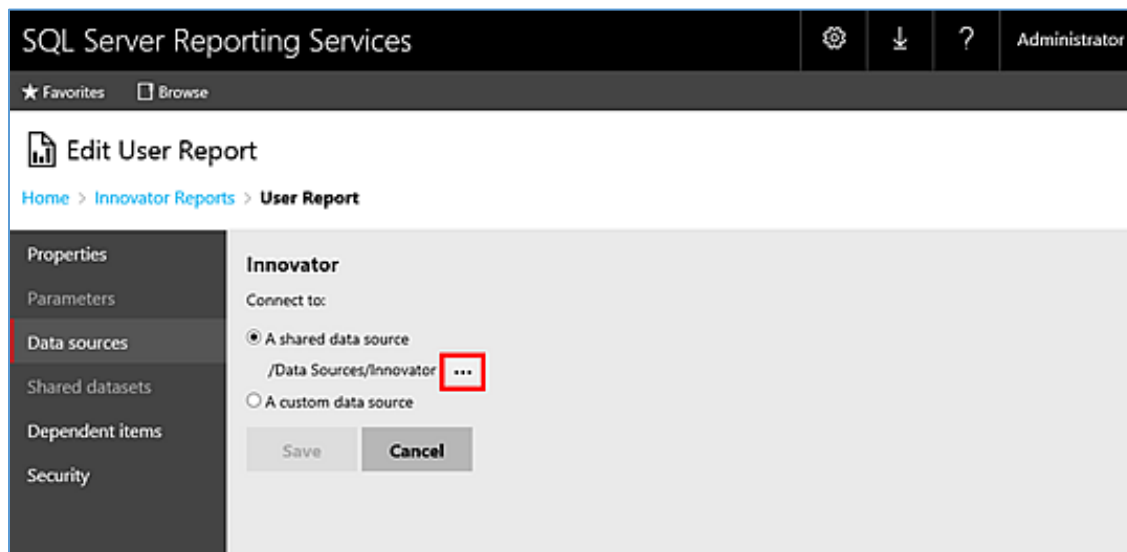


Figure 57.

4. If the named Shared Data Source is incorrect for this ReportServer, click ... to open the **Shared Data Source page**, navigate the Folders to locate the desired Shared Data Source, select it, and click **OK**.

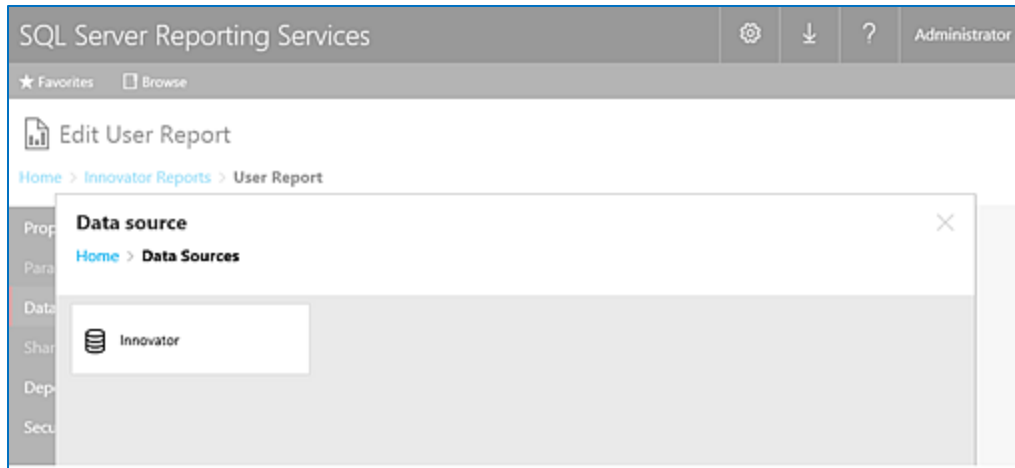


Figure 58.

8 Date Related SQL Queries

AML is the language of Aras Innovator, but some external add-ons do connect to Aras Innovator on a direct SQL level. Because Aras Innovator stores DateTimes in SQL Server as UTC, we have provided an add-on function for Microsoft SQL Server that allows the easy conversion of DateTime to and from UTC.

To convert DateTime values from UTC to a specific time zone, use the `ConvertToLocal({value},{Time Zone})` function. The time zone should be specified according to the registry key name of the time zone desired; refer to *Aras Innovator - Configuring Internationalization*. To use the `DEFAULT` parameter for the `{Time Zone}`, it is required that a value be set for `CorporateTimeZone`.

Example:

```
select item_number, created_on
from innovator.Document
```

Would be written as:

```
select item_number, innovator.ConvertToLocal(created_on,DEFAULT) as
CreadtedOn
from innovator.Document
```

Or:

```
select item_number, innovator.ConvertToLocal(created_on,'Eastern Standard
Time') as CreadtedOn
from innovator.Document
```

To convert DateTime values from a specific time zone to UTC, use the `ConvertFromLocal({value},{Time Zone})` function. The time zone should be specified according to the registry key name of the time zone desired; refer to *Aras Innovator - Configuring Internationalization*. To use the `DEFAULT` parameter for `{Time Zone}`, it is required that a value be set for `CorporateTimeZone`.

Example:

```
update innovator.Document
set effective_date = '1/1/2007 00:00:00.00'
```

Would be written as:

```
update innovator.Document
set effective_date = innovator.ConvertFromLocal('1/1/2007
00:00:00.00',DEFAULT)
```

Or:

```
update innovator.Document
set effective_date = innovator.ConvertFromLocal('1/1/2007
00:00:00.00','Eastern Standard Time')
```

This allows users to write queries in a familiar time zone context.

```
select item_number, innovator.ConvertToLocal(release_date, DEFAULT) as  
ReleaseDate  
  
from innovator.Document  
  
where created_on >= innovator.ConvertFromLocal('1/1/2007  
00:00:00.00', DEFAULT)
```